

The autonomy equation: How agentic AI reshapes trust and workload in routine productivity applications

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ABSTRACT

User experience and trust in AI-assisted technologies are key factors in controlling their adoption. We investigate these aspects in an Agentic AI platform that integrates routine productivity services and exhibits different levels of autonomy: a manual baseline that lacks AI-driven automation, an Agentic AI with medium autonomy that requires user confirmation before acting, and an Agentic AI with high autonomy that acts proactively for low-stakes tasks. The study, involving 230 participants with heterogeneous professional backgrounds, examines how autonomy of the system affects user activity, user workload, perceived support, and trust.

We found that both Agentic AI systems outperformed the baseline in user productivity. In task execution, they achieved a precision of over 82%, higher than the baseline's 65%. The recall of the Agentic AI system with high autonomy was 63%. This denotes much higher throughput than the system without AI-driven automation (14%). The Agentic AI systems outperformed the baseline in workload reduction (NASA-TLX Aggregate score) with a statistically significant difference. Both AI-driven systems received equivalent or slightly higher trust than the baseline. However, the system with medium autonomy was the best at balancing productivity gains and user preferences for control. Specifically, the correlations between individual user characteristics (Desirability of Control and Propensity to Trust) and the resulting trust in the systems suggest that the influence of personal traits on system evaluation is least pronounced when automation is combined with explicit user intervention. These results encourage the adoption of user-controllable Agentic AI architectures in multitasking support.

1. Introduction

Working environments expose people to a high level of multitasking. However, the explosion of digital media has contributed to the growth and ubiquity of this phenomenon, adding daily activities such as driving and managing chats to the parallel threads we carry out. Moreover, [Beuckels et al. \(2021\)](#) reported that media multitasking has become increasingly popular in the past decade, exacerbating this phenomenon.

Psychological science documented the behavioral cost of task switching, which induces errors and longer completion times. [Madore and Wagner \(2019\)](#) summarized the point as “subtle ‘switching’ costs cut efficiency, raise risk”. [Chen et al. \(2022\)](#) analyzed

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the interaction effects of interruption, considering the complexity of tasks and fatigue. Mark et al. (2016) reported the results of an in-situ experiment involving forty information workers whose online activity was tracked over two work weeks. They found that shorter focus duration on individual tasks (induced by task-switching) was associated with lower assessed productivity at the end of the day. Subardjo and Khan (2025) analyzed the consequences on personal health and productivity at work, highlighting the need for evidence-based tactics to reduce the negative effects of multitasking in today's digital world.

Research addressing productivity challenges in working environments has explored the support for multitasking, specifically focusing on interruption management (Alkahtani et al., 2020). Text summarization tools have also been introduced to help users manage dynamic and flexible workflows that involve them. For example, Read AI¹ can be connected to other applications using Webhooks to help people summarize their mailboxes and add events to their calendars. However, as these tools lack autonomy in executing actions, they require manual intervention. They also poorly address the cognitive challenges that professionals face because of continuous multitasking and context switching.

The emergence of Large Language Models (LLMs) has transformed how we envision AI assistance in modern work environments by making automation user-friendly. Autonomous agents have been a research focus in Artificial Intelligence for decades with virtual personal assistants as one of their most promising applications (Tulshan & Dhage, 2019). However, with the recent rise of powerful foundation models (Bommasani et al., 2022), LLM-based agents can now perceive the environment, make decisions, and take actions autonomously while maintaining friendly conversational interfaces with users (Wang et al., 2024). They can process Natural Language requests, understand implicit task requirements, and execute actions across multiple applications with contextual continuity (Xi et al., 2025). Agentic AI architectures powered by LLMs are promising solutions for application integration and multitasking support. Nevertheless, their probabilistic nature causes reliability issues. Hallucinations and uncertainty in their output (Chan et al., 2023) induce concerns about their suitability for high-stakes tasks. Thus, LLM-powered assistants raise questions about user trust, system autonomy, and human-AI collaboration patterns. Afroogh et al. (2024) point out that trust and distrust in AI serve as regulators. As they impact user acceptance, they could significantly control the level of diffusion of AI-assisted technologies.

Previous research has attempted to answer these trust issues with mixed results. Hoff and Bashir (2015) explained that user control on automated operations influences trust in the system. Qian and Wexler (2024) found that, in a software programming task, expert participants were more likely to reject AI-assistance than novices, showing that context is another influencing factor. However, He et al. (2025b) found that users can easily mistrust LLM agents working as daily assistants when they produce plausible plans in tasks with diverse levels of risk. Furthermore, Logg et al. (2019) hypothesized “algorithm appreciation”. They found people prefer algorithmic to human judgment in diverse decision-making domains, despite previous research hypothesizing the contrary. Similarly, Candrian and Scherer (2022) found that people prefer to delegate decisions to AI, compared to human agents, especially when decisions entail losses.

Most of the work focuses on closed decision-making tasks, such as solving a problem or buying a product, rather than analyzing human-AI interaction in software platforms that integrate multiple applications. Moreover, it mainly overlooks the impact of personality traits on user trust in AI-based automated assistants. As an exception, Hegner et al. (2019) related trust and concern about handing over control to an autonomous vehicle with people's cognitive and affective perception of its innovation and usefulness, as well as their Personal Innovativeness (Midgley & Dowling, 1978) and Locus of Control (Rotter, 1954)² traits.

In this work, we focus on multitasking. In this context, we investigate whether and how varying levels of autonomy in Agentic AI systems (corresponding to lower user control) affect trust and performance in routine productivity services. Specifically, we define a conceptual framework to examine the impact that different levels of system autonomy have on user activity, user workload, perceived support, and trust in the system. We also investigate whether Desirability of Control (Burger & Cooper, 1979) and Propensity to Trust (Körber, 2019; Mayer et al., 1995) moderate trust in Agentic AI. Unlike Hegner et al. (2019), we consider these personality traits because they describe users' attitudes towards giving control to external entities, rather than assessing where the control is located (as in Locus of Control).

By autonomy we mean the ability of a system to perform tasks derived from humans without specific human intervention, denoted as artificial autonomy in Hu et al. (2021). For our investigation, we pose the following research questions:

- RQ1: Does the autonomy level of an Agentic AI system improve user activity when managing multiple productivity services?
- RQ2: Does the autonomy level of an Agentic AI system reduce user workload when managing multiple productivity services?
- RQ3: Does the autonomy level of an Agentic AI system improve user perceived support when managing multiple productivity services?
- RQ4: Do different levels of AI autonomy impact user trust in the system?
- RQ5: Does Desirability of Control moderate the impact of different levels of AI autonomy on user trust in the system?
- RQ6: Does Propensity to Trust systems moderate the impact of different levels of AI autonomy on user trust in the system?

We organized a user study to answer our research questions. For this purpose, we designed an Agentic AI architecture that supports the integration of routine productivity applications in an open environment. Based on this architecture, we developed an email management platform integrated with a set of productivity apps for organizing users' calendars and to-do lists.

Parasuraman et al. (2000) defines 10 levels of automation, leading to an increasing autonomy, from a system that offers no assistance to one that decides everything, acts autonomously, and ignores the human. In our platform, we modeled three levels of autonomy in AI-driven automation:

¹ <https://www.read.ai/integrations>

² Locus of Control describes the degree to which people believe that they, as opposed to external forces, have control over the outcome of events in their lives.

1. A baseline system that integrates the productivity services but requires manual management of actions. In Parasuraman et al. (2000)'s model, this corresponds to level 1 ("the computer offers no assistance: human must take all decisions and actions").
2. An Agentic AI with medium autonomy that generates draft actions that users can review before performing them (level 5: the system suggests one alternative and "executes that suggestion if the user approves").
3. An Agentic AI with high autonomy that proactively executes low-stakes actions based on inferred user intent, providing users with awareness about its actions (level 9: the system "informs the human only if it, the computer, decides to"). For high-stakes actions, Agentic AI requires user confirmation.³

All systems provide awareness of the conflicts between the events reported in incoming emails and users' calendars. However, the two AI-driven systems (systems 2 and 3 above) also generate counterproposals to support the identification of viable solutions.

In the user study, we examined the impact of system autonomy on user trust and task performance in a simulated working context. We involved in the experiment 230 people with heterogeneous professional backgrounds. Our findings reveal that Agentic AI systems significantly improved task throughput and reduced user workload compared to the baseline system. Moreover, the highly autonomous system achieved the best results in these dimensions. Notice that increasing automation did not diminish user trust: participants trusted both Agentic AI systems equally or slightly more than the baseline. However, the system with medium autonomy provided an optimal balance between productivity gains and user preferences for control (Desirability of Control and Propensity to Trust). Specifically, the correlations between these personality traits and trust suggest that their influence on system evaluation is least pronounced when automation is combined with explicit user intervention. These results encourage the adoption of user-controllable Agentic AI architectures in multitasking support. In summary, our work brings three main contributions:

- The design and evaluation of an Agentic AI architecture that integrates multiple productivity services with three levels of system autonomy, ranging from manual management of actions to high AI-driven automation with awareness support.
- An investigation of how the Desirability of Control and Propensity to Trust moderate trust in Agentic AI systems exhibiting different levels of autonomy. This contributes to trust in automation theory. It shows that user characteristics impact trust differently depending on the AI's autonomy level, extending previous models (Hoff & Bashir, 2015).
- Empirical evidence from a comprehensive user study (n = 230) demonstrates that different levels of Agentic AI autonomy positively impact user workload, performance, and trust in AI-driven multitasking environments. This challenges findings that increasing automation can erode expert users' trust in AI (Qian & Wexler, 2024). Moreover, it provides new insights into this domain, as previous work has been carried out in high-stakes domains such as Autonomous Vehicles.

The remainder of this article is organized as follows: Section 2 positions our work in related research. Section 3 describes the user interfaces and functions offered by the three systems we developed. Section 4 presents our Agentic AI architecture and its communication flow. Section 5 describes the experimental design and the hypotheses we formulated. Section 6 outlines the results of our experiments, and Section 7 highlights the implications of our work. Section 8 reports the limitations, and Section 9 concludes the paper and summarizes our future work.

The code, prompts, and data are available at the following links: <https://github.com/ist-haac/agentic-ai-analysis> (data analysis) and <https://github.com/ist-haac/agentic-ai-app> (test application for our user study).

2. Literature review

In this section, we outline the background and related work related to Agentic AI architectures, Trust in AI, and Human-AI collaboration.

2.1. Background

Agentic AI leverages the conversational and problem-solving capabilities of Large Language Models (LLMs) to develop intelligent services that can operate with different levels of autonomy. These services assist users in various domains, such as information exploration (Trichopoulos et al., 2023), social sciences (Xu et al., 2024), public health interventions (Jo et al., 2023), hospitality management (Wang & Zhang, 2025), ontology development automation (Val-Calvo et al., 2025), task automation (Wu et al., 2024), and teamwork (Zheng et al., 2023).

LLMs are used as central controllers to design autonomous agents with humanlike decision-making capabilities. Equipped with external tools, they can generate and perform plans autonomously (Wang et al., 2024). However, Agentic AI is far from perfect. Firstly, the autonomous planning ability of LLMs is limited (Valmeekam et al., 2023). Moreover, Agentic AI poses potential risks due to LLMs' hallucinations and bias (Chan et al., 2023), given the probabilistic nature of neural networks and their limitations in handling sparse data. Although the factuality of LLMs tends to improve with their size (Tam et al., 2023), the problem remains unsolved. Hallucinations can also lead to misinterpreting users' requests. This raises dependability issues because systems can operate differently from expected. Furthermore, the limited size of text that can be prompted to an LLM prevents context management in extended conversations, resulting in a suboptimal user experience (Kim & Choudhury, 2021).

These issues raise trust concerns to be addressed. According to Afroogh et al. (2024), "trust and distrust in AI serve as regulators and could significantly control the level of this diffusion, as trust can increase, and distrust may reduce the rate of adoption of AI."

³ Parasuraman et al. (2000)'s 10th level, which represents black-box autonomous systems, is out of the scope of this work.

Indeed, the evolved user expectations in the post-LLM era show that users increasingly view AI as a collaborative partner rather than a mere tool, reshaping trust calibration processes (Lindgren, 2025).

From a human-centered AI (Shneiderman, 2022) perspective, Torkamaan et al. (2024) argued that integrating LLMs into socio-technical systems poses further complex challenges due to the intricate interplay between societal structures, human behavior, and technological innovation. Similarly, as AI assistants become more humanlike and exhibit greater agency, discerning when user trust is justified requires consideration not only of competence but also alignment between the competing interests, values, or incentives of AI assistants, developers, and users (Manzini et al., 2024). Li et al. (2025) identified interaction, ethics, and evolution (to address the fulfillment of human needs in diverse and dynamic environments) as key research domains for Human-centered Intelligent Systems. Cao et al. (2025) proposed a classification of key elements of group decision-making in social networks (trust models, internal structure, and consensus mechanism) for AI and LLMs. They emphasized the critical relationship between trust and Large Language Models in that context: trust mechanisms must be carefully designed to account for the characteristics of LLM-based decision-making systems, including their probabilistic nature, potential for hallucination, and the need for transparent and explainable AI processes.

These and other issues raise concerns about delegating high-stakes decisions and tasks to AI systems (Kaur et al., 2022). A relevant amount of work is ongoing to:

1. Improve the reliability of this technology by extending the software architectures to contrast issues such as memory limitations and algorithmic bias (see Section 2.2).
2. Improve user trust by enhancing human-AI interaction through explainability and by explicitly involving users in problem-solving rather than delegating this activity to black-box AI agents (see Section 2.3).

2.2. Agentic AI architectures

Various architectures enhance Agentic AI's reliability by extending LLMs with specialized modules that assist in fulfilling specific roles, managing long-term memory, and operating based on the surrounding context. See Wang et al. (2024) for a survey on this topic. Retrieval-augmented generation (RAG) techniques (Lewis et al., 2020) are used to retrieve external information, enhancing the consistency and factual accuracy of generated responses. Furthermore, Chain-of-Thought prompting (Wei et al., 2022) improves context feeding in prompts. This enables LLMs to articulate their reasoning process explicitly by breaking complex tasks into intermediate steps. For example, Seo et al. (2025) used this technique to address long-term recall and personalization problems. They incorporated short/long-term memory components into intelligent assistants to enhance context awareness and response coherence.

We propose an Agentic AI architecture that integrates external APIs, specialized internal reasoning modules, and LLMs to support users in managing multiple productivity services. Our architecture differs from those described in Wang et al. (2024) because a deterministic controller (rather than an LLM) manages the interaction with the user for predictability. The controller integrates specialized reasoners to manage functions such as conflict detection and long-term context management. In contrast, it delegates creative tasks, such as email creation, to an LLM. The controller also supports system extensibility. Our architecture orchestrates and synchronizes components and the LLM across different interaction sessions using contextual information in a loosely coupled approach.

Some frameworks, such as Microsoft AutoGen⁴ and IBM's BeeAI⁵ help non-technical users develop specialized AI agents and integrate them into multi-agent architectures through prompt engineering. In comparison, our architecture supports the integration of LLMs with traditional software to leverage their capabilities and compensate for their limitations. It also maintains a long-term context that addresses the recognized limitations of LLMs in context management.

2.3. Trust in AI and human-AI collaboration

Trust is described as a positive expectation an agent has about the behavior of other agents from a subjective perspective. For example, Mayer et al. (1995) describes it as “the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor”.

2.3.1. Trust and reliance

The literature about trust in automation established that trust is a major predictor of reliance on AI advice (Kohn et al., 2021; Schaefer et al., 2016). User trust and reliance on AI emerged as relevant issues since early research on automation to prevent harm in the use of autonomous systems. Currently, interest in such aspects has grown dramatically because AI and algorithmic decision-making impact our daily lives (Kaur et al., 2022), supporting task execution in domains with different levels of stake. For example, Klingbeil et al. (2024) reported an overreliance on AI advice for financially risky decisions in an experiment. Moreover, Qian and Wexler (2024) found that, in a software programming task, participants of all expertise levels increasingly relied on an AI assistant (Google Bard), despite reporting lower trust in AI. Furthermore, in a study on critical thinking in knowledge work, Hao-Ping et al. (2025) found that “while Generative AI can improve worker efficiency, it can inhibit critical engagement with work and can potentially lead to long-term overreliance on the tool and diminished skill for independent problem-solving.” To address these issues, Brachman et al.

⁴ <https://microsoft.github.io/autogen/0.2/>

⁵ <https://www.ibm.com/think/news/beeai-open-source-multiagent>

Table 1

Overview of the related work on trust and control of AI assistance.

Authors	Analyzed dimensions	Personality traits	User control	Number of levels of system autonomy
Hoff and Bashir (2015)	trust in automation	Big5 Extroversion and Neuroticism	Binary: yes/no	1
Schaap et al. (2024)	algorithmic aversion	–	Binary: yes/no	1
Tsai and Brusilovsky (2019)	exploration effort	–	yes (tuning)	1
Brusilovsky (2023)	learner control in AI collaboration	–	yes (personalized content selection, open user models)	1
Lawless et al. (2024)	interactive decision support	–	yes (criticism-based)	1
Peng et al. (2025)	interactive decision support	–	yes (open user models)	1
He et al. (2025b)	interactive planning support	–	Binary: yes/no	1
Hegner et al. (2019)	Autonomous Vehicles adoption	Locus of Control, Sensation seeking, Personal innovativeness	no	1
Our present work	AI assistants for productivity management	Desirability of Control, Propensity to trust	Binary: yes/no	3

(2025) highlighted the “need for further transparency in Agentic AI systems to support end-users in evaluating system outputs and help them build effective mental models”.

Previous work has attempted to calibrate trust and the reliance on AI by extending AI recommendations with model-centric explanations of results (Morrison et al., 2024). Wang and Ding (2024) found explainable AI can improve human decision accuracy and behavioral trust. However, an imperfect explainable AI can lead to misleading explanations (Sadeghi et al., 2024; Zhang et al., 2020). Moreover, He et al. (2025b) observed that enhanced conversations powered by LLM agents amplified overreliance. They postulated that a possible cause of this phenomenon is the illusion of explanatory depth. Explanations are thus desirable to improve AI acceptance, but must be carefully planned to avoid misleading users (Cau et al., 2023; Cau & Spano, 2025). Kim et al. (2024) found that using Natural Language expressions of uncertainty may reduce overreliance on Large Language Models. Swaroop et al. (2025) proposed to personalize decision-making assistance to different users and tasks for improving human-AI team performance.

Our work builds on these findings and aims to reduce the cognitive load on the user while managing multiple tasks. Thus, we propose a concise interaction model in which the Agentic AI system explains its behavior through summaries and short notifications, depending on its level of autonomy. This approach enables us to manage both process transparency (explaining the generation of counterproposals to address conflicts) and outcome transparency (communicating what actions the system has taken or will take through appropriate notifications).

2.3.2. Trust and user control on AI assistance

Table 1 outlines the research on the relationship between trust and user control in human-automation interaction. Hoff and Bashir (2015) explained that trust is a layered concept that includes dispositional, situational, and learned trust. The former depends on the individual user; the other two are based on experience with the system. They highlighted that the user’s control over automated operations influences trust. Moreover, they reported that previous work found some associations between the “Neuroticism” and “Extroversion” Big5 personality traits, and dispositional trust. Schaap et al. (2024) investigated algorithmic aversion (a tendency to prefer human decisions to machine-generated ones) while controlling increased benefits and decreased user control. In a survey on scenarios that concern financial or sentimental choices, they found that “users accept or reject decisions and decisional agents based on their predicted benefits and the ability to exercise control over the decision.”

The recommender systems research investigated user control and empowering models, e.g., by enabling users to tune the impact of different algorithms in a hybrid recommender system (Tsai & Brusilovsky, 2019), balance learner and system control in AI in education through personalized content selection and open user models (Brusilovsky, 2023), provide multiple recommendation to increase user choice (Fink et al., 2024), and elicit preference information in conversational approaches to decision-making (Lawless et al., 2024). These interaction models are related to human-AI collaboration, which has been studied since Horvitz (1999)’s work on mixed-initiative interaction. Horvitz’s goal was to improve Human-Computer Interaction by coupling automated services and direct manipulation in user interfaces.

In Agentic AI, human-AI collaboration is proposed to address various open issues. Peng et al. (2025) proposed addressing inaccurate intent recognition in exploratory tasks by enabling users to correct the system’s hypotheses on their information needs (we classify this approach as open user models, albeit the authors do not use this terminology). Moreover, He et al. (2025b) applied plan-and-solve prompting (Wang et al., 2023) to allow users to revise LLM-generated plans in a daily assistant AI agent.

Our work differs from previous ones in the following aspects:

- We investigate people’s reactions to the provision or lack of control in transparent systems. In contrast, previous work focused on steering system behavior through direct manipulation, correction of system behavior, or tuning the impact of algorithms. However, it analyzed black-box decision makers.

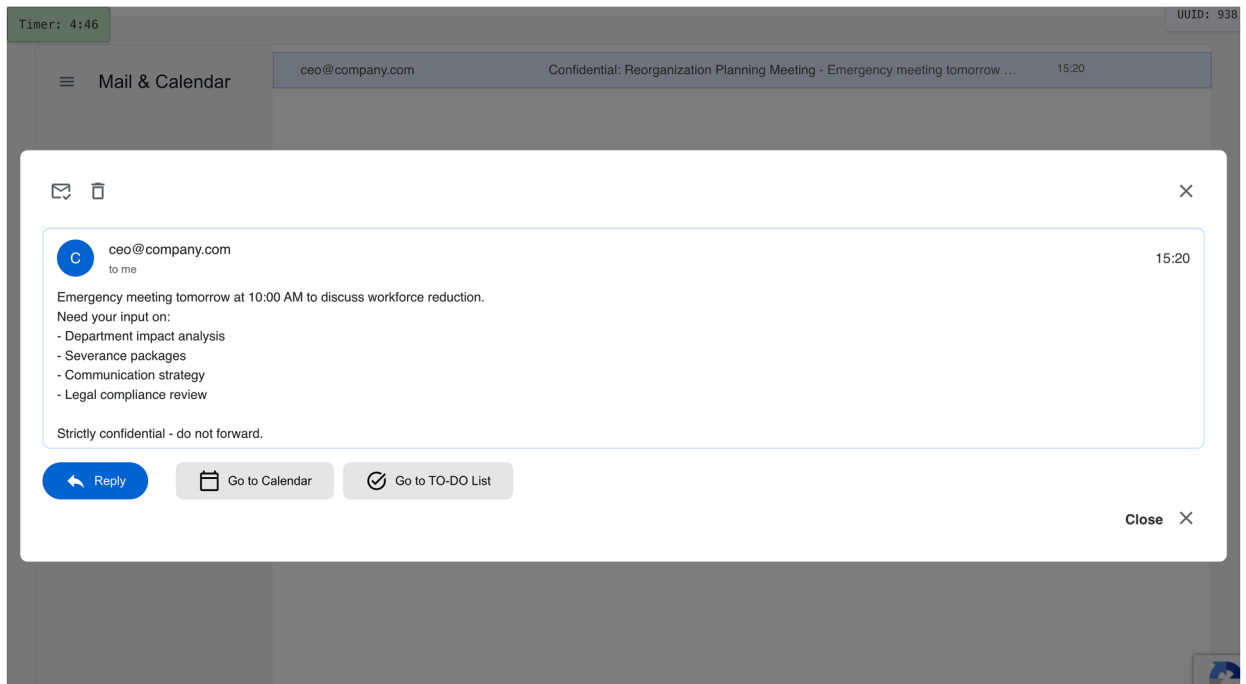


Fig. 1. SYS0. Baseline system to manage productivity services without automation.

- We study user interaction with systems that exhibit varying levels of autonomy: (i) no autonomy, where the user is in full control because the system relies on their manual intervention to perform basic actions; (ii) middle autonomy, where an Agentic AI system suggests actionable operations giving the user full control on whether enacting them or not; (iii) high autonomy, where the Agentic AI system performs low-stakes actions autonomously (but provides awareness support) and requires user confirmation for high-stakes ones. In comparison, previous work overlooked user experience with systems that exhibit different levels of autonomy.

We investigate the moderating effects of two personality traits related to user control and trust (Desirability of Control (Burger & Cooper, 1979) and Propensity to trust (Körber, 2019)) to examine user interaction with Agentic AI systems having medium or high autonomy. Specifically, we analyze the correlation between individual user characteristics and trust in automation to achieve a fine-grained picture of the relationships between these dimensions. Previously, correlations have been studied in the high-stakes field of Autonomous Vehicles (AV). In a survey, Hegner et al. (2019) found that “trust in the technology and the concern about handing over control to a machine go hand in hand as respondents’ cognitive and affective perception of this innovation”. Perceived usefulness is another influential factor. Furthermore, based on previous findings that Locus of Control and Sensation Seeking influence the acceptance of Autonomous Vehicles, these authors found that Personal Innovativeness (Midgley & Dowling, 1978) positively influences the intention to adopt an AV. Our work is different because it targets the context of AI assistants to support daily workflow management in productivity environments, with substantially different risks and emotional profiles.

3. Systems

This section outlines the user interfaces and functions offered by the three systems we developed for our experiments. We designed these interfaces inspired by widely used mail clients, with standard email viewing panes, action buttons, and navigation elements that participants could recognize from their daily work routines. This was aimed at reducing cognitive overhead and aligning the simulated scenario with real-world tasks. In all user interfaces, the left side displays the timer and buttons to switch between productivity services. For example, see component “Mail & Calendar” in the background of Figs. 2 and 3.

Systems differ in level of autonomy and in the user interfaces for email management. Table 2 summarizes the key features of these systems to guide their presentation. Notice that although these systems integrate specific services (i.e., an email application, a to-do manager, and a calendar manager), the underlying Agentic AI architecture is open and can be extended with other apps.

3.1. System without AI-driven automation - SYS0 (baseline)

Fig. 1 shows SYS0, which offers calendar and to-do management functions. According to Parasuraman et al. (2000)’s scale, the level of initiative of this system is 1. Users must manually process incoming messages, calendar events, and to-do items without assistance. However, the email viewer provides basic management controls to support switching between services. The email view

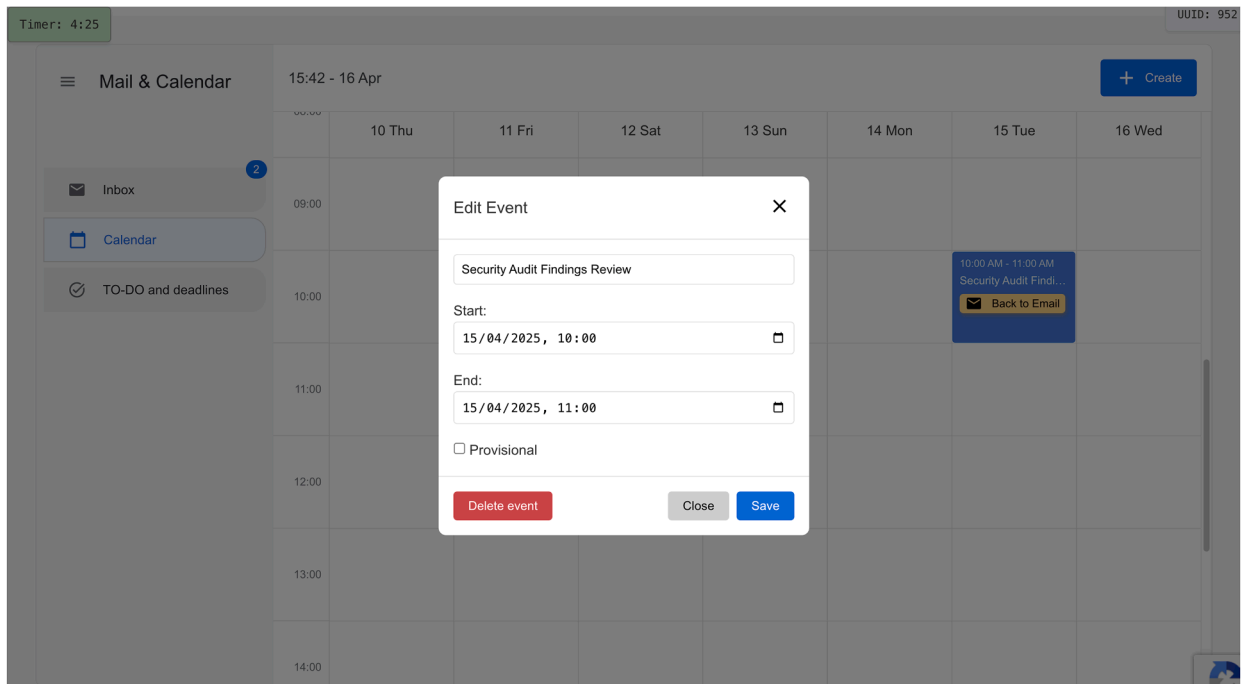


Fig. 2. Our application for managing calendar events.

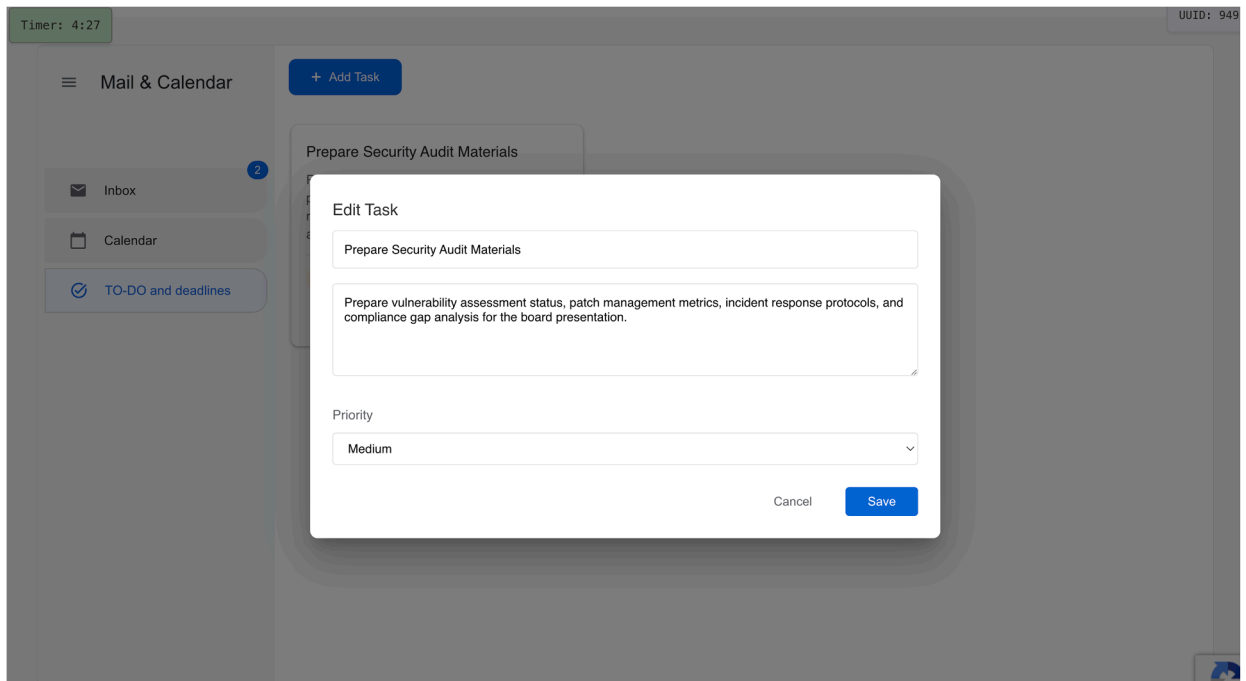


Fig. 3. Our application for managing to-dos.

displays action buttons for replying to the email, manually adding the event to the calendar (“Go to Calendar”), and manually creating a to-do item (“Go to TO-DO List”). By clicking on these buttons, productivity applications open automatically. Thus, the user can directly add, edit, or delete an existing item within a unified environment (Figs. 2 and 3). As it lacks autonomy, SYS0 does not explicitly inform the user about calendar conflicts. However, it provides basic awareness support by coupling overlapping events in the affected calendar cells, similar to what standard calendar apps do. Thus, the user can identify problematic events.

Table 2

Comparative overview of SYS0, SYS1, and SYS2, considering domain-independent features and specific functions offered to users. We measure the system's level of autonomy according to [Parasuraman et al. \(2000\)](#)'s scale. ○ indicates that the user controls decision-making and action execution. △ denotes system control in decision-making and action execution.

High-level features	SYS0 (Baseline)	SYS1 (Middle Autonomy)	SYS2 (High Autonomy)
System's Level of Autonomy	1 (User performs actions)	5 (User approves all system suggestions)	9 (User approves high-stakes actions, system autonomously executes the other ones)
Awareness support	Visualization of the state of productivity apps	Visualization of the state of productivity apps + action suggestions	Visualization of the state of productivity apps + draft emails + notifications
System response to user modifications	System applies changes to modified items	System applies changes in cascade	System applies changes in cascade
Conflict Detection	Visualization of overlapping events in calendar	Notifications of conflicts and counterproposals	Notifications of conflicts and counterproposals
AI-Driven Assistance	No	Yes	Yes
Specific functions			
Email Draft Composition	○	△ + ○	△ + ○
Calendar Management	○	△ + ○	△ (upon-email-send)
To-Do List Management	○	△ + ○	△ (upon-email-send)

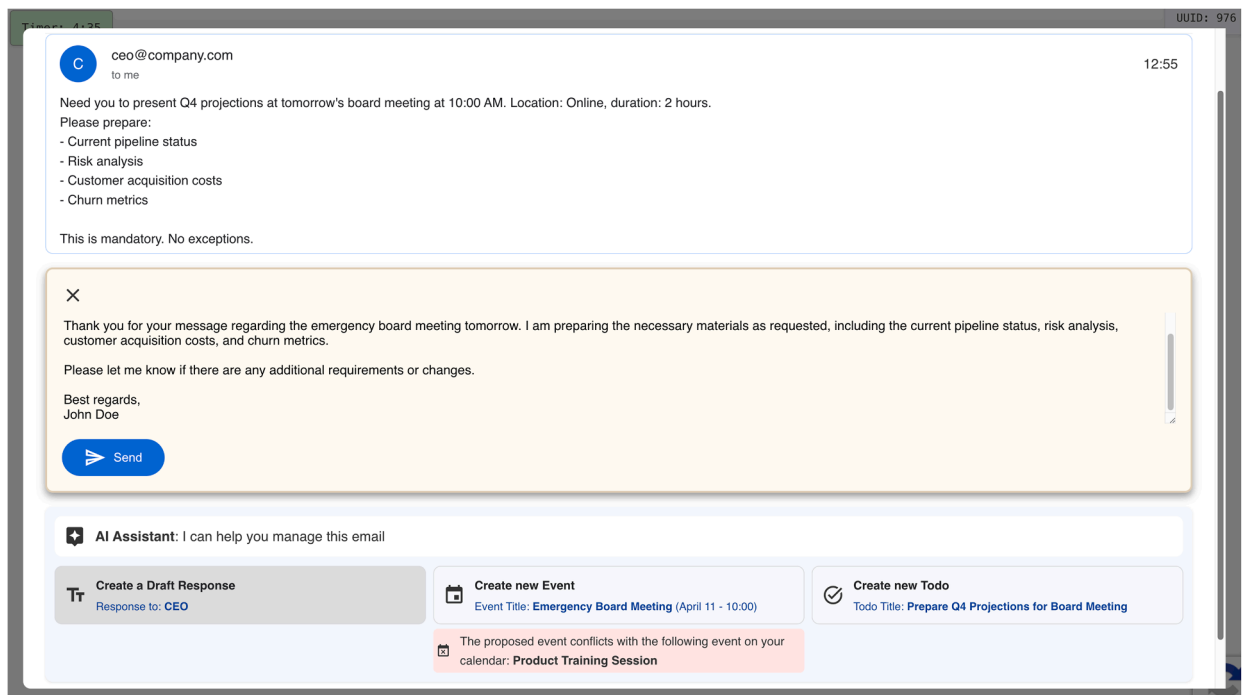


Fig. 4. SYS1. System for managing productivity services with medium AI-driven autonomy. User interface with notification of calendar conflict.

3.2. System with medium AI-driven autonomy - SYS1

[Fig. 4](#) shows the user interface of SYS1, whose level of initiative in [Parasuraman et al. \(2000\)](#)'s scale is 5. This system assists the user through an AI agent ("AI Assistant: I can help you manage this email"). The system analyzes incoming messages and extracts data from their sender, topic, body, and date to generate action suggestions across multiple productivity services.

SYS1 simplifies action execution and provides awareness support, but it requires the user's confirmation for all actions. It presents the suggestions to the user through buttons in the lower portion of the user interface. The set of actions that can be suggested is: "Create a Draft Response", "Create new Event"/"Update Event"/"Delete Event", "Create new To-do"/"Update To-do"/"Delete To-do". All buttons display details about the actions they trigger (e.g., date and time of an event to be scheduled) for transparency.

SYS1 has a conflict detection service to monitor the user's calendar. If an incoming message proposes an event that overlaps with the user's existing commitments, the system attempts to find an alternative date for the event that is compatible with the user's

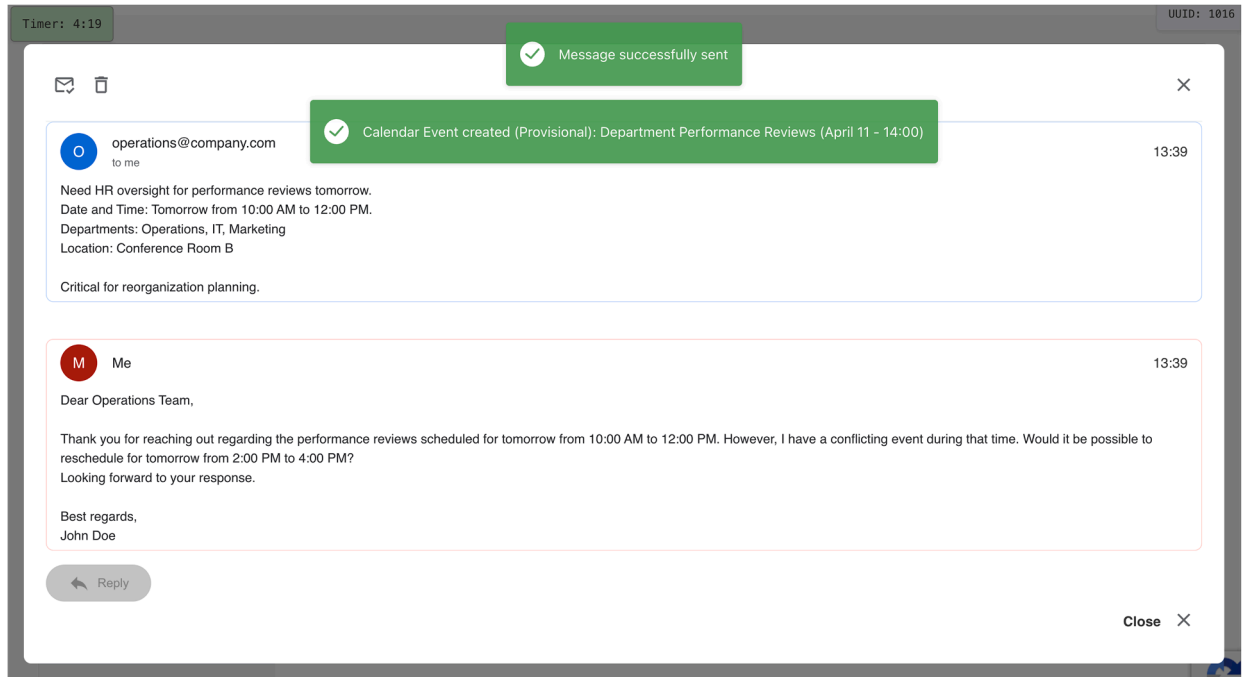


Fig. 5. SYS2. System for managing productivity services with high AI-driven autonomy. Sample counterproposal.

calendar. If it succeeds, it generates a counterproposal to help the user solve the conflict. Otherwise, it maintains the date proposed in the message. In both cases, it notifies the user about the details of the conflict through a notification.

Fig. 4 shows a step of the interaction with SYS1. The higher portion of the user interface presents an incoming message that the user has opened. The message sets a meeting, causing a conflict in the user's calendar. In this example, the system has not found an alternative date for the requested meeting. Thus, the "Create new Event" button proposes to set the event on the original date. However, the system displays a pink notification: "The proposed event conflicts with the following event on your calendar: Product Training Session". In the figure, the system has generated a draft message in the yellow widget because the user has clicked the "Create a Draft Response" button (disabled to show that the action has been performed). The user can edit the draft and finally send it to the recipient by clicking the blue button ("Send").

SYS1 monitors the changes in the proposed actions to capture the user's intent and update the impacted action suggestions. For example, if the user modifies the draft message proposing an alternative meeting time and sends it as a response, the system refreshes the calendar and to-do management buttons in cascade. In this way, suggested actions are up-to-date and ready to be applied without modifications.

3.3. System with high AI-driven autonomy - SYS2

Fig. 5 shows the user interface of SYS2, which offers the same functions as SYS1 but has a different level of autonomy (9 in Parasuraman et al. (2000)'s scale). This system autonomously enacts low-stakes actions, but requires user confirmation for high-stakes ones.

When the user opens an email message, SYS2 proactively creates a draft response, possibly including a counterproposal to address calendar conflicts, and shows it in the user interface. However, as emails represent a possibly high-stakes type of action, the system enables the user to update the draft as needed, and finally send it by clicking the "Send" button.

When the user sends an email response, its content is considered as confirmed and can be safely executed. Thus, SYS2 autonomously creates the corresponding entries in cascade in the integrated productivity services, using the message content to set items (to-dos, calendar events) accordingly. Furthermore, it displays notifications to inform the user that it has completed such actions on her/his behalf.

Fig. 5 shows the user interface of SYS2 immediately after the user has clicked the "Reply" button (now disabled) to send a response message. As the incoming email proposed an event in conflict with the user's previous commitments, the response message mentions the problem and includes a counterproposal (move the event to 2 PM or 4 PM). The green notifications inform the user that the system has delivered the message and added the event ("Department Performance Reviews") to the calendar. However, the event has been added to the calendar as "provisional" at 2 PM ("Calendar Event created (Provisional): ...") to highlight that it differs from the meeting requested in the incoming message (10 AM). This is aimed to make the user aware that the event is not definitive and could change again, if the recipient makes a further counterproposal.

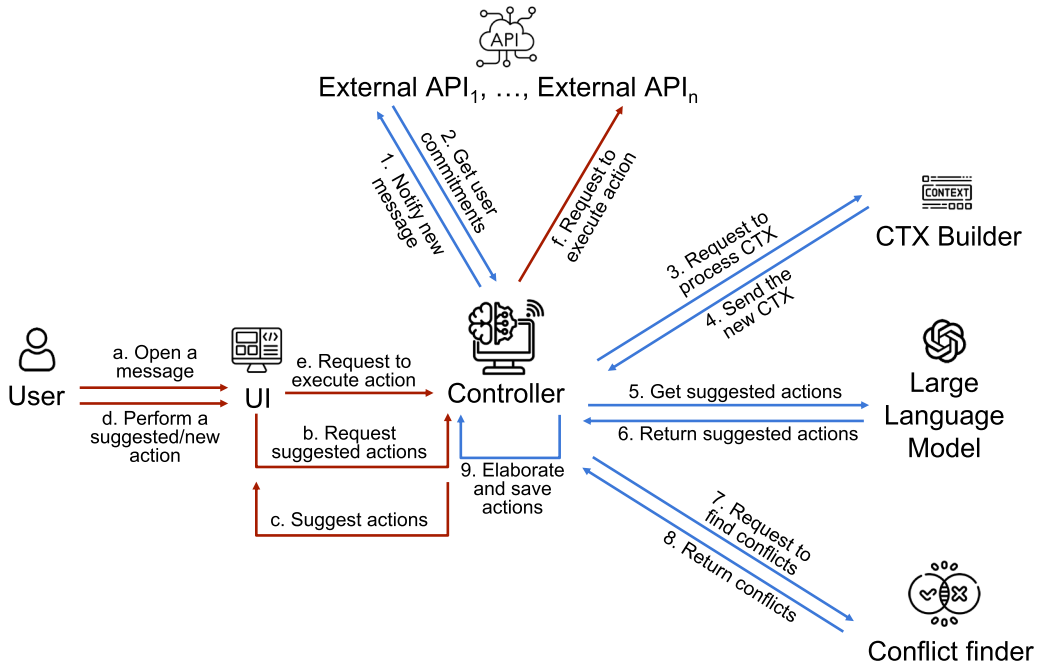


Fig. 6. Architectural framework and communication flows of the proposed Agentic AI system.

4. Agentic architecture and communication flow

Fig. 6 presents the Agentic AI architecture underlying SYS1 and SYS2 and the communication flow among the system components. We overlook the architecture of SYS0, which integrates external APIs of productivity services without using intelligent components. In our Agentic AI architecture:

- The User Interface (UI) manages the communication channel with the user. It makes the system a unified intelligent environment that supports email, calendar, and task management. See Section 3.
- The Controller manages user (OAuth) authentication and single sign-on in the platform. It orchestrates the other system components. It prompts the LLM to process user input and retrieve action suggestions, enacts the actions that the LLM suggests by invoking external APIs, and sends the responses to the UI. To reduce response latency, the Controller prompts the LLM as soon as the mail client receives a new message. However, the user's calendar may change due to other events occupying certain time slots before the message m is processed by the user (by performing some actions on it). To ensure that the action suggestions remain up to date when the user processes m , the Controller re-invokes the LLM to refresh the system suggestions.
- An LLM elaborates prompts and suggests actions in response to incoming emails. In our use case, the actions are drafting an email response, creating/updating/deleting a calendar event, or creating/updating/deleting a to-do item.
- The CTX Builder creates the user context that the LLM uses to generate action suggestions, based on the email in focus, the user's inbox content, calendar events, and existing to-dos. However, Large Language Models have difficulty understanding numbers and have limited temporal reasoning capabilities (Yang et al., 2025). To address these issues, the CTX builder converts the calendar information, which only indicates the user's busy slots, to a timeline that also reports the free slots. The Controller then injects the context information (in JSON format) into the prompt for the LLM.
- The Conflict Finder takes as input the user's calendar events and the new events derived from the incoming email. It checks for overlaps, and returns the identified conflicts. These checks are useful for two reasons: (i) incoming emails might propose events that overlap with the user's previous commitments; (ii) LLMs sometimes hallucinate and suggest conflicting events that represent false positives; thus, their output must be verified.
- External APIs integrate productivity services. Besides those we consider in this work, other ones could be added by connecting the Controller to External APIs.

Our Agentic AI architecture works according to a workflow inspired by the one described by NVIDIA⁶ and IBM⁷:

1. The Controller manages the sensing phase. The environment to be sensed consists of the user (through the UI) and the integrated productivity services (through APIs).

⁶ <https://blogs.nvidia.com/blog/what-is-agentic-ai/>

⁷ <https://www.ibm.com/think/topics/agentic-ai-vs-generative-ai>

2. The Conflict Finder and the LLM manage the reasoning phase. During this phase, the system generates a list of actions (e.g., suggesting draft responses, managing the user's calendar, or the to-do list) to react to the received input.
3. The Controller orchestrates sensing and actuation of actions in the environment.
4. The CTX Builder supports the learning phase by maintaining the user context to react to events that occur in the environment. Currently, this module has limited learning capabilities and works as a memory module for the LLM to maintain long-term user information. However, it can be expanded to infer information for personalization, such as, learning users' daily routines to personalize calendar management (Berry et al., 2011).

The system processes emails through two communication flows:

- The primary flow is depicted by blue arrows with numbers. When a new email arrives, the External API notifies the Controller (step 1), which retrieves user commitments from the External API (step 2) and requests context processing to the Context Builder (step 3). The Context Builder structures this information, incorporating the new message alongside existing user commitments. It returns the updated context (step 4) to the Controller, which then requests action suggestions from the LLM (steps 5–6). Then, the Conflict Finder module checks the suggestions (steps 7–8). Finally, the Controller elaborates and saves the proposed actions (step 9).
- The secondary flow (red arrows with characters) is user-initiated and starts when a user opens a message in the UI (step a). The UI requests suggested actions from the Controller (steps b–c), allowing the user to select and perform one of these actions or initiate a new one (step d). In the system with medium autonomy (SYS1), users select the actions to be performed by clicking the AI assistant's buttons. In contrast, the highly automated system (SYS2) shows the draft response in the UI. When the user clicks the "Send" button, the system delivers the message and automatically executes the suggested actions impacted by the message (e.g., adding an event to the calendar). Moreover, it notifies the user that the actions have been executed for awareness purposes.

In both SYS1 and SYS2, if the user modifies the draft email, the agent updates the actions impacted by the message accordingly. After the user action (sending an email, clicking a button, etc.), the UI requests to execute the action (step e), which the Controller delegates to external APIs (step f).

Our Agentic architecture is agnostic to the LLM that can be seamlessly updated to exploit technological advances and improved performance. Moreover, it supports the integration of additional productivity applications by extending the Controller API and the orchestration workflow. Managing the user context across productivity services facilitates integration with diverse applications and communication platforms. This approach also supports the management of orchestration aspects concerning action dependencies across applications, such as setting tasks for users through a shared document (Goy et al., 2017).

5. Experimental design

This section describes the methodological aspects of our user study, which has been approved by the Ethics Committee of the University of Torino (Protocol Number: 0681861).

5.1. Hypotheses

The experiment is designed to investigate the impact of varying levels of AI-driven automation on human interaction with a system that integrates routine productivity services. We aim to explore how autonomy influences users' trust and performance. To answer the research questions outlined in Section 1, we formulate four hypotheses:

HP1: Enhancing the autonomy of Agentic AI for productivity service integration improves user activity.

LLM-based agents can perceive the environment, understand user intent and implicit task requirements, and make decisions to steer action execution across multiple applications (Wang et al., 2024; Xi et al., 2025). This capability can be used in AI-driven autonomous systems to streamline task execution and reduce manual intervention. In this perspective, consistent with Qian and Wexler (2024)'s results in software programming, we expect that as the level of autonomy increases from SYS0 to SYS2, user activity will be higher.

HP2: Enhancing the autonomy of Agentic AI for productivity service integration reduces workload.

As discussed in Section 1, psychological science documented the behavioral cost of task switching, which induces errors, fatigue, and reduced productivity (Chen et al., 2022; Madore & Wagner, 2019; Mark et al., 2016). Leveraging AI-driven autonomous systems to streamline action execution can be exploited to contrast the negative effects of multitasking. Thus, similar to Qian and Wexler (2024)'s results, we expect that as the level of autonomy increases from SYS0 to SYS2, workload will decrease.

HP3: Enhancing the autonomy of Agentic AI for productivity service integration increases user perceived support.

LLM-based systems have advanced conversational interfaces and context-aware action execution capabilities (Wang et al., 2024; Xi et al., 2025). As such, they can be employed in AI-driven autonomous systems to interpret users' intents and act on users' behalf, supporting their work proactively. In our experiments, the autonomous and interaction capabilities of SYS1 and SYS2 are based on an LLM, while SYS0 is manual. Thus, we postulate that user-perceived support will be higher in the former systems than in the latter. For this hypothesis, we overlook SYS0 because it lacks AI-driven support.

HP4: Enhancing the autonomy of Agentic AI for productivity service integration negatively impacts user trust in the system.

As reported in Section 2, trust seems to depend on the degree of control the user has over automated operations (Hoff & Bashir, 2015). Moreover, user expertise might lead reject AI-assistance in some domains, such as software programming

(Qian & Wexler, 2024). As higher levels of automation are inherently more complex and potentially more opaque to operators, they may engender less trust (Lewis et al., 2018). Furthermore, the observability of automation varies considerably across different levels of autonomy (Lee & See, 2004). Information acquisition and analysis automation remain relatively observable, whereas decision selection and action implementation automation (as in SYS2) are less observable when not being directly monitored. This reduced observability impairs users' ability to form accurate mental models of system behavior, hindering trust calibration. Finally, psychological reactance theory suggests that when people experience a threat to or loss of their free behaviors, they experience an unpleasant motivational arousal that motivates them to restore their freedom (Steindl et al., 2015). As higher system autonomy transfers control from the user to the system, it can trigger reactance and reduce trust.

Based on these premises, we posit that user trust in AI systems decreases if they are more autonomous in managing productivity tasks. In other words, we expect to observe higher trust scores for SYS0 compared to SYS1 and SYS2, and possibly the lowest trust in SYS2, which has the highest level of autonomy.

To refine the perspective provided by *HP4*, we investigate personality traits as a further source of influence on user trust in autonomous systems. Hegner et al. (2019) reported an observed relationship between Locus of Control and the acceptance of autonomous functions in Autonomous Vehicles. Moreover, they analyzed emotional aspects that impact trust in the adoption of such vehicles. We believe that other key aspects in a productivity environment are whether users *want* to give control to an automated system, and how strongly they *are inclined* to trust automation. Thus, we analyze the relationship between user trust and their Desirability for Control (Burger & Cooper, 1979), or Propensity to Trust (Körber, 2019; Mayer et al., 1995). The next two hypotheses focus on such aspects.

HP5: The trust in Agentic AI for integrating productivity services with different levels of autonomy negatively correlates with the Desirability of Control.

The Desirability of Control describes an individual's motivation to exert influence over their life events and environment. This personality trait is closely related to psychological reactance theory, which posits that individuals experience an unpleasant motivational arousal when they perceive a threat to or loss of their free behaviors (Steindl et al., 2015). Higher levels of system autonomy transfer decision-making authority from the user to the system. Thus, they might be perceived as constraining users' behavioral freedom, particularly for individuals with high Desirability of Control who are motivated to maintain influence over their environment. Furthermore, as discussed in *HP4*, higher automation levels reduce system observability and create opacity in decision-making processes (Lee & See, 2004). For users with a high Desirability of Control, this opacity may be problematic because it removes direct control and limits their ability to understand and predict system behavior, compounding the perceived threat to their autonomy.

On this basis, we postulate that users with a high Desirability for Control are less inclined to trust systems that automate tasks without their direct input. As the level of autonomy increases from SYS0 to SYS2, we expect to see a stronger negative correlation between the Desirability of Control and trust in such systems.

HP6: The trust in Agentic AI for integrating productivity services with different levels of autonomy correlates with Propensity to Trust systems.

Propensity to Trust is a generalized expectation about the trustworthiness of others. We thus study whether an individual's tendency to trust technology is correlated with trust in AI-driven autonomous systems. Lee and See (2004) state that "a small difference in the predisposition to trust may have a substantial effect when it influences an initial decision to engage the automation". This suggests the need for further investigation on how the propensity to trust may influence trust in an automated system.

We hypothesize that people who tend to trust technology more will also exhibit greater trust in AI-driven productivity systems, regardless of the level of autonomy of such systems.

5.2. Study design

We employed a between-subjects experimental design with three conditions corresponding to the systems described in Section 3: SYS0 (baseline without AI-driven automation), SYS1 (AI-driven automation with medium autonomy), and SYS2 (AI-driven automation with high system autonomy). Sample size was determined *a priori* using G*Power with the following parameters: effect size $f = 0.25$ (low-medium), $\alpha = 0.01$, and power = 0.8, yielding a target sample of 228 participants.

We used previous expertise with routine productivity applications as an inclusion criterion for participating in our user study. This is relevant to prevent biases in results caused by the observed moderating effect of user knowledge (Wang & Sun, 2025).

The experiment was guided by a test application that randomly assigned participants to one of the three conditions and administered questionnaires. The application logged the user's actions during interaction with the assigned system, monitoring the time to perform activities. The test application administered 3 well-known, validated questionnaires (Desirability of Control (Burger & Cooper, 1979), Trust in Automation (Körber, 2019), and NASA-TLX (NASA, 2019)) to elicit information concerning personal characteristics and effort while interacting with SYS0, SYS1, and SYS2. Moreover, it administered a few questions we defined to elicit specific information about the participant's opinion on AI assistance in e-mail calendar and to-do management.

Table 3
Sample structure for managing an email thread.

Field	Value
Job Category	sales
Scenario	sales_0
Thread's initial Email	Email ID: EMAIL_1 Sender: training@company.com Email Subject: Product Training Session Email Body: ‘Mandatory product update training tomorrow at 10:00 AM. Duration: 2 h Location: Online All sales staff must attend as this covers critical new features.’
Follow-up	Follow-up ID: FOLLOW_1 Option A: RESCHEDULE_EVENT ‘If the user agrees to participate in the ‘Product Training Session’, you MUST RESCHEDULE the EVENT to a DIFFERENT datetime.’ Option B: ACCEPT_EVENT ‘If the user cannot attend the ‘Product Training Session’ or proposes to reschedule the ‘Product Training Session’, you MUST ACCEPT the proposed new datetime.’ Option C: GENERAL_ACKNOWLEDGMENT ‘If the user input does not match the default types, a response confirmation is sent.’

5.3. Participants

We recruited participants through Prolific,⁸ a platform for online behavioral research. Appendix A in the Appendix reports the instructions given to Prolific workers before accepting to partake in the study.

The description of our experiment on Prolific briefly presented the study and its goals. To ensure validity and domain-specific knowledge, we set a filter to recruit professionals across 10 job categories: Customer Experience, Product Management, Data Analysis, Design/Creative, Research, Finance/Accounting, Legal, Human Resources, Information Technology, and Sales. This sample allowed us to evaluate systems across various professional contexts where workers commonly manage emails and use productivity services.

Section 6.1 describes the demographics of participants.

5.4. Data (task scenarios)

To ensure a controlled user study, we developed a set of pre-compiled high-pressure scenarios covering each of the 10 job categories. Every scenario includes 8 to 10 email threads (8.3 on average) initiated by a fixed email by a simulated actor, such as the CEO of a company. The actor interacted with the participant recruited for the experiment via email. In each thread, the initial (pre-compiled) email sets a meeting or asks the user to create to-dos and send responses within a limited timeframe. As the participant might respond to the initial email, the thread often includes a follow-up pattern to prompt the simulated actor to react. The requests and events reported in the mails sent by the simulated actor are contextualized in the 10 job categories selected for the experiment and matched to participants’ jobs.

To manage email threads, our test application used a Large Language Model that operated as the simulated actor. This LLM is not part of our Agentic AI architecture; we briefly describe the simulated actor in Section 5.5.

In this experiment, we designed the follow-up patterns to pressure participants. For instance, if the initial email sets a meeting and the user accepts it, the follow-up is typically a request to reschedule the meeting rather than closing the email thread. However, we wanted to prevent cyclic negotiation turns in this experiment. Thus, if the user responds to the initial email with a counterproposal (e.g., suggesting a different schedule), the follow-up accepts it. Table 3 shows the structure of an email thread belonging to the “sales” job category:

- The initial email for this thread is “EMAIL_1” from the simulated actor “training@company.com”. The “Email Body” field contains the text of the email.
- The field “Follow-up” specifies how the simulated actor should respond to the user’s answer to “EMAIL_1”. The A, B, and C options of “FOLLOW_1” specify the alternative directives that can be applied, depending on the user’s answer to “EMAIL_1”. For instance, Option A specifies that, if the participant agrees to join the “Product Training Session” event, the simulated actor should apply the “RESCHEDULE_EVENT” directive. The options are described in Natural Language because they represent prompts for the LLM (simulated actor) that generates the body of the follow-up messages.

⁸ <https://www.prolific.com/>

Table 4

Sample prompt for generating dynamic follow-up emails from the simulated sender.

Follow-up condition	Prompt
RESCHEDULE_EVENT	<pre>## RULES FOR 'RESCHEDULE_EVENT' TYPE: <rules> 1. Knowing that NOW is {ctx.get('current_date')} and it is {ctx.get('current_time')}, FIND the DATE and TIME of the EVENT. To do this, you MUST consider the 'content' of the EMAIL THREAD. 2. IMPORTANT: The EVENT'S DATE and TIME is WITHIN the message content of the EMAIL THREAD. 3. STRONGLY AVOID TO CONSIDER the messages 'timestamp'. 4. The generated response MUST mandatory have a DATETIME DIFFERENT from that contained in the EMAIL THREAD message. 5. The response MUST reschedule the event to a DIFFERENT DATE and DIFFERENT TIME than the original one. 6. The response MUST reschedule the event to a RANDOM DATE BETWEEN THOSE FOUND IN 'current_week_days'. 7. The response MUST CONTAIN the new date. 8. The response MUST CONTAIN the new datetime, namely the day of the week, the calendar date, and the time. </rules></pre>

Table 4 shows the specification of the “RESCHEDULE_EVENT” directive, i.e., the instructions for proposing a rescheduling of the initially planned event. The instructions are expressed as a Natural Language prompt for the LLM.

We generated the scenarios using the sonar model by Perplexity as an LLM.⁹ For the follow-ups we designed five options (RESCHEDULE_EVENT, ACCEPT_EVENT, CANCEL_EVENT, ADD_TASK, GENERAL_ACKNOWLEDGMENT) and we asked the LLM to generate the follow-ups using a subset of the most coherent options. Then, three researchers manually refined and validated them to ensure realism, appropriate complexity, and domain-specific terminology.

The complete set of scenarios is available at <https://github.com/ist-haac/agent-ai-analysis/blob/main/data/scenarios.json>. The prompts specifying the implementation of directives are available at the following link: https://github.com/ist-haac/agent-ai-app/blob/main/server/llm_prompts/followup_prompt.py. Note that the specification of how to implement directives (e.g., the strategy for rescheduling a meeting) is abstract and does not depend on the email thread.

5.5. Procedure

The experiment followed a structured procedure consisting of the sequential steps described below:

1. Participants reviewed and agreed to the informed consent, which was mandatory to continue the experiment. The text of the informed consent is available here: <http://bit.ly/44rcDvQ>.
2. Participants received a detailed description of the study and goals, including its focus on productivity application management.
3. The test application collected participants' nicknames, which were used to personalize the interface. The application also administered:
 - The Desirability of Control questionnaire, which measures individual differences in the propensity to control events (Burger & Cooper, 1979);
 - The questions of the “Propensity to Trust” construct, a subscale of the Trust in Automation questionnaire (Körber, 2019). This construct does not concern the experience with a specific system. It measures an individual's tendency or willingness to trust others. According to Körber (2019), it determines the other dimensions of trust measured by the overall questionnaire. The construct consists of the following statements on a 5-Point Likert scale:
 - (a) One should be careful with unfamiliar automated systems.
 - (b) I rather trust a system than I mistrust it.
 - (c) Automated systems generally work well.
 While this scale was originally developed for traditional automated systems, it has also been used in LLM-powered systems (Govers et al., 2025; He et al., 2025a,b).
4. Participants viewed annotated screenshots of their assigned system (SYS0, SYS1, or SYS2) with detailed explanations of its features and functionalities. Moreover, the test application assigned each participant to a scenario matching the user's job category provided by Prolific. This approach ensured people would encounter familiar professional jargon and typical work situations within their domain of expertise, making email management realistic and engaging.
5. Participants engaged in a 5-min simulation using the assigned system to manage emails, calendar events, and tasks. During the simulation, people were involved in the (8 to 10) email threads belonging to the assigned scenario (see Section 5.4). As previously

⁹ This model was available at the time of writing: <https://www.perplexity.ai/api-platform/resources/meet-new-sonar>.

Table 5
Summary of the dimensions and measures selected for analyzing our hypotheses.

Hypotheses	Dimension	Measures
HP1	User Activity	Number of actions performed by users Message response rate Precision and Recall of user actions
HP2	Workload	NASA-TLX questionnaire (NASA, 2019)
HP3	Perceived Support	Questionnaire on perceived support from AI-driven systems Open feedback from participants (<i>Time pressure and increased productivity</i>)
HP4	Trust in Agentic AI	Trust in Automation (TiA) Questionnaire (Körber, 2019): - Reliability/Competence - Understanding/Predictability - Familiarity - Intention Of Developers - Trust In Automation
HP5	Desirability of Control	Desirability of Control questionnaire (Burger & Cooper, 1979)
	Trust in Agentic AI	Trust in Automation (TiA) questionnaire (Körber, 2019): - Reliability/Competence - Understanding/Predictability - Familiarity - Intention Of Developers - Trust In Automation Level of autonomy of the Agentic AI Open feedback from participants (<i>Automation and User Control</i>)
HP6	Propensity to Trust	Trust in Automation (TiA) questionnaire (Körber, 2019): - Propensity to Trust
	Trust in Agentic AI	Trust in Automation (TiA) questionnaire (Burger & Cooper, 1979): - Reliability/Competence - Understanding/Predictability - Familiarity - Intention Of Developers - Trust In Automation Level of autonomy of the Agentic AI

described, the emails required to perform actions such as: (i) responding to the sender (simulated actor) with the appropriate information; (ii) creating calendar events for meetings set by the email; (iii) adding items to a to-do list based on tasks mentioned in the email; (iv) managing conflicts between existing commitments and new requests.

Participants were instructed to process as many emails as possible while maintaining accuracy. The user interface included a clock to display elapsing time during the experiment. The test application closed the interaction with the system automatically when the time limit expired. Note that the experiment was intentionally short to press participants and force them to decide which commitments could be accepted or shifted in time under a stressful condition. This was aimed at assessing how effectively the systems supported users in managing multiple tasks within a constrained time interval.

6. After the interaction with the assigned system, participants completed the post-task questionnaires:
 - Trust in Automation: in this phase, the test application administered the questions determining dimensions Reliability/Competence, Understanding/Predictability, Familiarity, Intention of Developers¹⁰, and Trust in Automation. We remind the reader that the questions determining the dimension Propensity to Trust were administered in step 3 above.
 - NASA-TLX questionnaire, which assesses perceived workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration.
 - Moreover, participants in the SYS1 and SYS2 conditions completed a Likert scale questionnaire evaluating AI assistance for email, calendar, and to-do management.
7. Finally, participants provided open-ended feedback about their experience with the system they used.

To manage email threads during the experiment, the test application used the *sonar* model by Perplexity, which operated as the simulated sender of the messages (simulated actor). To generate follow-ups, our test application prompted the LLM with the initial email of the thread, the participant's answer to such email, the follow-up conditions (Table 3), and the specifications of directives (Table 4). Based on the interpretation of the user's answer, the LLM selected the directive and applied it to generate the follow-up email. The test application received the email from the LLM and injected it into the email client of our Agentic AI architecture.

5.6. Data collection and analysis

We collected objective performance metrics and subjective experience measures:

- **Performance Metrics:** Number of actions performed (email answered, calendar events created/updated/deleted, to-dos created/updated); message response rate; precision and recall of actions, comparing user actions against expert-defined correct actions for each email.
- **User Profiling Metrics:** Scores from (i) the Desirability of Control questionnaire, and (ii) the Propensity to Trust subscale of the Trust in Automation questionnaire.
- **Post-task Questionnaires:** NASA-TLX questionnaire (Table 8); the remaining subscales from the Trust in Automation questionnaire (Table 10); the ad-hoc questionnaire (Table 9).

For statistical analysis, we used non-parametric tests (Kruskal-Wallis for omnibus comparisons across the three systems, Mann-Whitney U for pairwise comparisons, and non-parametric TOST based on Mann-Whitney U to test for equivalence) due to non-

¹⁰ This dimension assesses whether users believe that the developers have designed the automation with the user's best interests in mind, are acting ethically, and are not seeking to exploit or harm users. The Trust in Automation questionnaire is available here: https://github.com/moritzkoerber/TiA_Trust_in_Automation_Questionnaire/blob/master/1_manual/TiA_Manual_Eng.pdf.

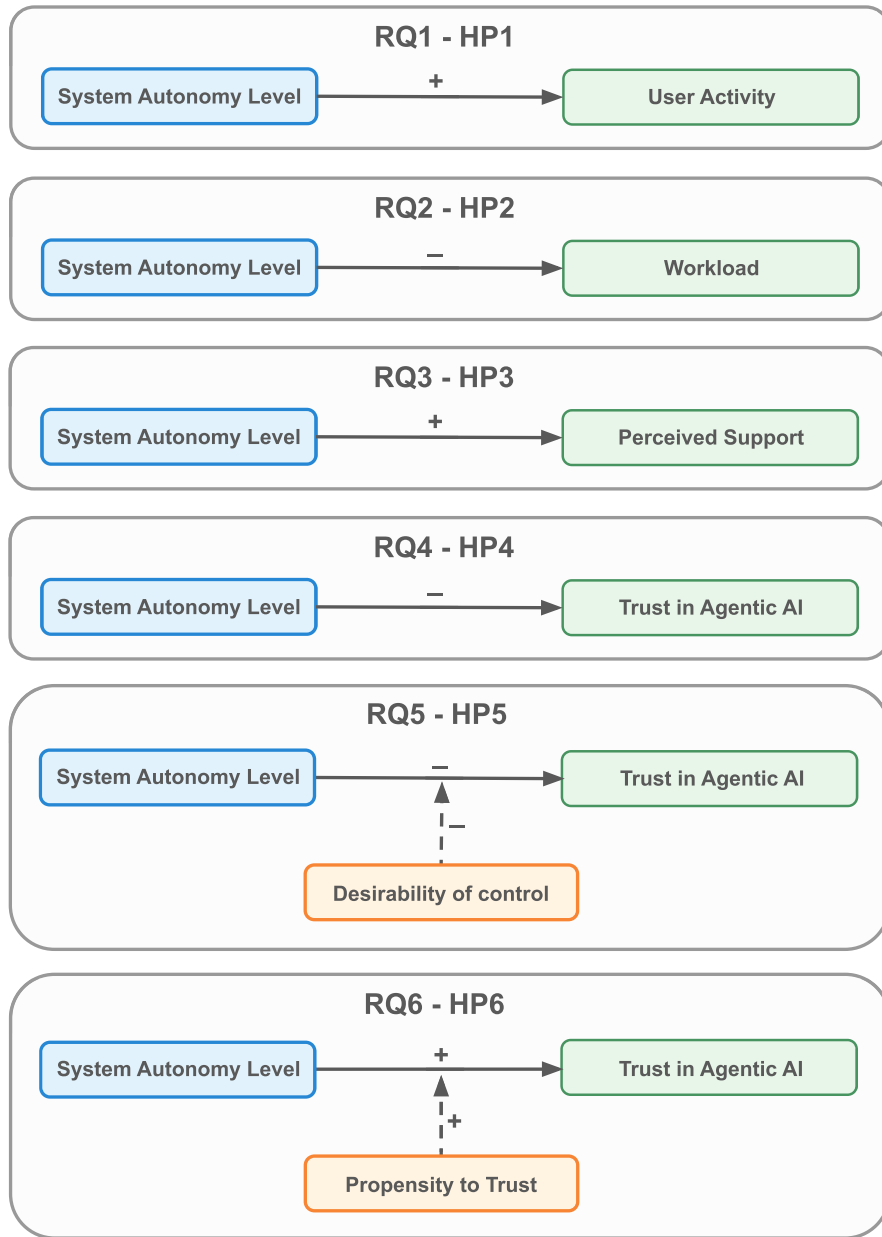


Fig. 7. Conceptual framework representing the relationships between independent, moderating, and dependent variables, according to our research hypotheses. We use the following color code: blue for independent variables, orange for moderating ones, and green for dependent variables. The dashed arrows link moderating variables to the affected dependencies. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

normal distributions in our data. The TOST test measures statistical equivalence; thus, a low p-value indicates statistical equivalence between the compared conditions (Lakens, 2017). The TOST test requires a threshold to be set, below which it is believed two series can be considered equivalent (Lakens, 2017). We set this to 10 for the NASA-TLX questionnaire, and 0.5 for everything else. This corresponds to a 10% variation of the respective scales.

We adjusted p-values for multiple comparisons of all tests using the Bonferroni-Holm method to control for family-wise error rate. This is the overall probability of incorrectly detecting significance in at least one comparison when multiple tests are performed.

To assess the practical significance of our findings, we reported standardized effect sizes alongside p-values: Cohen's d for pairwise comparisons (Mann-Whitney U tests) and η^2 (eta squared) for omnibus tests (Kruskal-Wallis). Effect sizes are interpreted following standard conventions: for Cohen's d , $|d| < 0.2$ indicates negligible effects, $0.2 \leq |d| < 0.5$ small effects, $0.5 \leq |d| < 0.8$

Table 6

User activity levels for SYS0, SYS1, and SYS2. We report Mean values and Standard Deviation. The best values are in boldface.

	SYS0	SYS1	SYS2
Number of actions	6.60(4.48)	18.22(9.38)	24.92(8.09)
Normalized message response rate	0.30(0.21)	0.66(0.31)	0.86(0.25)

medium effects, and $|d| \geq 0.8$ large effects; for η^2 , values < 0.01 indicate negligible effects, $0.01 - 0.06$ small effects, $0.06 - 0.14$ medium effects, and ≥ 0.14 large effects. Additionally, we conducted post-hoc power analyses to assess the true statistical power achieved in our tests.

- **Qualitative Data:** Participants' responses to open-ended questions about user experience during interaction with the assigned system.

To examine the impact of personality traits on user trust in Agentic AI, we conducted correlation analyses between individual difference measures (Desirability of Control, Propensity to Trust) and post-task evaluation measures. To establish the strength and direction of these relationships, we performed a simple linear regression, from which we derived Pearson correlation coefficients as the square root of the coefficient of determination (R^2). We report 95% confidence intervals for all correlation coefficients, to convey the precision of these relationships.

Table 5 summarizes the dimensions and measures we selected to validate our research hypotheses. Fig. 7 presents our conceptual framework graphically, considering our research hypotheses. The figure shows the relationships among our independent variable (System Autonomy Level), the moderating variables (Desirability of control and Propensity to trust), and the dependent variables (User Activity, Workload, Perceived Support, Trust in Agentic AI).

6. Results

This section outlines the results of the user study we carried out to test the hypothesis and answer the research questions.

6.1. Participant overview

Our participant pool consisted of 390 individuals. We screened out 160 participants who did not complete the study, had an incompatible device, or failed one or more attention checks. The resulting sample had size = 230. Then, we manually analyzed the interaction sessions of participants in this sample to verify that they interacted coherently with the assigned system. Specifically, we ensured that they replied to emails, staying on topic with the content of the incoming messages. We found rare incoherent messages, plausibly due to fatigue in managing the task under stressful conditions. Overall, there were 4 incoherent messages, written by different users, out of 230 for SYS0; 3 out of 668 for SYS1, and 2 out of 929 for SYS2. Thus, we decided to consider this sample as the final one for our study.

The 230 participants who completed the study were compensated at a rate of £6.06/hr for their time and effort. Concerning the data provided by Prolific:

- Gender distribution: 48% females, 51% males, 1% preferred not to answer.
- Age demographics: 22% of participants were between 18 and 25; 45% between 26 and 35; 20% between 36 and 45; 14% over 45.
- Geographic distribution: participants primarily came from Africa (38%), Europe (37%), North America (18%), Asia (4%), and South America (2%).
- Ethnic backgrounds: 45% White/Caucasian, 8% Asian, 39% Black/African, and 5% multiracial or other.
- While English was the primary language for 68% of participants, the study included speakers of 28 other languages, with Spanish (7%), German (5%), and French (4%) being the most frequent.

6.2. HP1. Enhancing the autonomy of agentic AI for productivity service integration improves user activity.

To address this hypothesis, we measured the message response rates and user actions in the SYS0, SYS1, and SYS2 conditions.

6.2.1. Normalized message response rate

The normalized message response rate, calculated as the ratio of replied messages to received messages during the 5-min simulation, revealed clear differences across systems. SYS2 achieved the highest response rate ($M = 0.86$, $SD = 0.25$), followed by SYS1 ($M = 0.66$, $SD = 0.31$) and SYS0 ($M = 0.30$, $SD = 0.21$). This confirms that both AI-driven systems substantially improved participants' ability to manage incoming messages compared to the baseline, with the most autonomous system (SYS2) showing the strongest effect.

Table 7

Precision and recall. The best values are in boldface. For the statistical significance, (*) $p < 0.05$, (**) $p < 0.01$, (***) $p < 0.001$ indicate the thresholds for the Kruskal-Wallis test. Complete pairwise results are available in [Appendix C](#).

	SYS0	SYS1	SYS2	Kruskal-Wallis p-value	η^2 (Power)
Precision	0.6462	0.8469	0.8220	0.043040275 (*)	0.025 (0.570)
Recall	0.1394	0.5058	0.6308	0.000000000 (***)	0.517 (1.000)

6.2.2. User actions

Concerning user actions, we first analyzed the average number performed during the simulation. Actions included event and to-do list management, and message replies, performed manually by the user (in SYS0), triggered by explicit actions (e.g., by clicking button “Create new Event” in SYS1) or automatically performed by the system as a consequence of the user’s actions (e.g., the calendar management actions enacted when the user sends a response email in SYS2). See [Table 6](#) for results.

Given its high level of autonomy, SYS2 generated the highest number of actions ($M = 24.92$, $SD = 8.09$) followed by SYS1 ($M = 18.22$, $SD = 9.38$) and the baseline SYS0 ($M = 6.60$, $SD = 4.48$). This data shows a clear trend of increased user activity in the AI-driven systems compared to the baseline.

6.2.3. Accuracy in action execution

We evaluated Precision and Recall on user actions to assess their accuracy; see [Table 7](#). We computed these measures on the list of actions performed by each participant during the 5-min simulation as follows:

- We manually annotated each user action as either right or wrong. We considered *wrong* the actions containing factual errors and/or inaccuracies in terms of date, time, or description compared to email discussions.
- Moreover, we computed the number of missing actions in the email threads. The instructions initially provided to participants specified that when an incoming message required them to participate in a meeting, an event, or perform a task, the user was expected to respond and add these items to the calendar/to-do list. Thus, we considered the commitments mentioned in the emails but not recorded in the appropriate organizational tools as *missing* actions.

Based on this data:

- We computed Precision as the ratio of correct user actions (*correct*) to the total number of actions recorded by the user in the Calendar and To-do list, using the formula $Precision = \frac{correct}{correct+wrong}$.
- Recall was derived as the ratio of correct actions to the total number of commitments that should have been recorded (*tot*): $Recall = \frac{correct}{tot}$.

As shown in the table, the precision of SYS1 and SYS2 was higher than 82%, while SYS0 had 65%. This confirms that AI-driven action execution support (either by clicking buttons in SYS1 or performed automatically when sending the draft in SYS2) reduced errors. Moreover, the recall for SYS0 (14%) is significantly lower than SYS1 (51%) and SYS2 (63%).

This shows that the support provided by the Agentic-AI systems increases accuracy.

6.2.4. Conclusion

This suggests that both middle and high system autonomy increased activity in the email management task, enabling users to accomplish more work within the same time interval, with higher accuracy.

HP1 is confirmed.

6.3. HP2. Enhancing the autonomy of agentic AI for productivity service integration reduces workload.

To address this hypothesis, we examined the workload perceived by users through the NASA-TLX questionnaire.

[Table 8](#) presents the results across all dimensions for SYS0, SYS1, and SYS2. [Fig. B.1](#) (in the Appendix) shows these results in a plot.

Focusing on the aggregate NASA-TLX score (overall workload, reported in the last row of the table), we found a significant main effect across systems ($p < .001$). Pairwise Mann-Whitney tests revealed that SYS0 imposed a significantly higher workload than SYS1 and SYS2. In contrast, the difference between SYS1 and SYS2 was not significant.

We now consider the individual questions of NASA-TLX. In SYS0, mental demand (question N1 in the table) was significantly higher than SYS1 and SYS2. This is also reflected in the fact that users felt more rushed (question N3) in SYS0 and, conversely, felt more successful (N4) in accomplishing the task in SYS1 and SYS2. SYS0 required users to work the hardest (N5), and mostly induced negative emotions (N6).

Another practical evidence of workload reduction is given by the participants’ actions on the user interface of SYS1 and SYS2. When these systems generated draft emails with counterproposals to resolve calendar conflicts, users accepted such proposals in 94.4% of times for SYS1 and 97.6% for SYS2. In this way, they could save time and effort in finding alternative dates for conflicting events.

HP2 is confirmed.

Table 8

NASA-TLX (NASA, 2019) Questionnaire Results comparing SYS0, SYS1, and SYS2. For each trust dimension, results show Mean scores with Standard Deviation (M (SD)), significant pairwise differences (Diff:), and equivalences (Eq:). We use $\diamond$ for the comparison with SYS0, $\dagger$ for SYS1, $\star$ for SYS2, with one/two/three symbols indicating $p < 0.05/p < 0.01/p < 0.001$. For example, in the SYS0 column, Diff. $\dagger\dagger$ indicates that the difference between SYS0 and SYS1 is statistically significant with $p < 0.01$. The fifth column shows the results of the Kruskal-Wallis test. We use the same thresholds of the other columns and denote them with $\star$. Complete pairwise results are available in [Appendix C](#).

Question	SYS0	SYS1	SYS2	Kruskal-Wallis p-value	η^2 (Power)
N1: How mentally demanding was the task?	64.48 (24.71)	45.58 (30.47)	37.70 (29.38)	0.000000179	0.128
	Diff: $\dagger\dagger, \star \star \star$ Eq: –	Diff: $\diamond\diamond$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	***	(1.000)
N2: How physically demanding was the task?	35.39 (30.29)	28.25 (32.50)	20.59 (27.80)	0.001979663	0.046
	Diff: $\star$ Eq: –	Diff: – Eq: –	Diff: $\diamond$ Eq: –	*	(0.851)
N3: How hurried or rushed was the pace of the task?	78.83 (23.42)	63.18 (31.06)	51.97 (32.15)	0.000000281	0.124
	Diff: $\dagger, \star \star \star$ Eq: –	Diff: $\diamond$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	***	(1.000)
N4: How successful were you in accomplishing what you were asked to do?	56.49 (28.56)	33.38 (28.65)	25.92 (21.90)	0.000000001	0.179
	Diff: $\dagger\dagger\dagger, \star \star \star$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	***	(1.000)
N5: How hard did you have to work to accomplish your level of performance?	62.40 (25.08)	48.05 (33.81)	36.32 (27.21)	0.000001026	0.113
	Diff: $\star \star \star$ Eq: –	Diff: – Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	***	(0.999)
N6: How insecure, discouraged, irritated, stressed, and annoyed were you?	48.31 (33.34)	26.17 (30.91)	28.36 (30.19)	0.000042071	0.080
	Diff: $\dagger\dagger\dagger, \star \star$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	Diff: $\diamond\diamond$ Eq: –	***	(0.984)
NASA-TLX Aggregate	55.49 (13.62)	46.31 (19.00)	41.50 (16.84)	0.000000348	0.122
	Diff: $\dagger, \star \star \star$ Eq: –	Diff: $\diamond$ Eq: –	Diff: $\diamond\diamond\diamond$ Eq: –	***	(1.000)

Table 9

Questionnaire on perceived support from SYS1 and SYS2. The thresholds are the same as in [Table 8](#).

Statements	SYS1	SYS2	Mann-Whitney p-value	d (Power)	TOST p-value
Q1: The proactive support in the Mail section improved my email management (e.g. drafting emails).	4.12 (0.91)	4.16 (0.86)	1.000000000	–0.046 (0.058)	0.000782862 (***)
Q2: The proactive support for Calendar management (e.g. automatically creating events by using the AI assistant) helped me schedule my events effectively.	4.31 (0.78)	4.01 (1.04)	1.000000000	0.322 (0.473)	0.215838508
Q3: The proactive support for Todo management (e.g. automatically creating tasks by using the AI assistant) helped me manage my tasks efficiently.	4.31 (0.71)	4.12 (0.90)	1.000000000	0.237 (0.285)	0.022965478 (*)

6.4. HP3. Enhancing the autonomy of agentic AI for productivity service integration increases user perceived support.

We measured perceived support across systems by exploiting two information sources. The former is the questionnaire on perceived support from AI-driven systems; the latter is the open feedback provided by participants.

[Table 9](#) shows the results of the post-test questionnaire administered to users assigned to SYS1 or SYS2. For a graphical representation of these results, see [Fig. B.2](#) in the Appendix.

The results show statistical equivalence between the two systems in perceived support for email (question Q1) and to-do management (Q3). The difference in the results of question Q2 (Calendar management support) is not statistically significant. However, both

Table 10

Trust in Automation (TiA) (Körber, 2019) Questionnaire Results for SYS0, SYS1, and SYS2. We use the following symbols: $\diamond$ for SYS0, $\dagger$ for SYS1, $\star$ for SYS2, and the same notation as in Table 8. Complete pairwise results are available in Appendix C.

Dimension	SYS0	SYS1	SYS2	Kruskal-Wallis p-value	η^2 (Power)
Reliability/Competence	3.26 (0.71) Diff: – Eq: –	3.50 (0.66) Diff: – Eq: $\star \star \star$	3.52 (0.71) Diff: – Eq: $\dagger \dagger \dagger$	0.392070888	0.025 (0.568)
Understanding/Predictability	2.81 (0.53) Diff: – Eq: $\dagger \dagger, \star \star$	2.69 (0.57) Diff: – Eq: $\diamond \diamond, \star \star \star$	2.63 (0.49) Diff: – Eq: $\diamond \diamond, \dagger \dagger \dagger$	1.000000000	0.012 (0.295)
Familiarity	3.12 (1.16) Diff: – Eq: –	3.00 (1.19) Diff: – Eq: –	2.97 (1.04) Diff: – Eq: –	1.000000000	0.000 (0.050)
Intention Of Developers	3.47 (0.72) Diff: – Eq: $\dagger, \star$	3.60 (0.70) Diff: – Eq: $\diamond, \star \star$	3.59 (0.81) Diff: – Eq: $\diamond, \dagger \dagger$	1.000000000	0.000 (0.050)
Trust In Automation	3.39 (0.97) Diff: – Eq: –	3.75 (0.83) Diff: – Eq: $\star$	3.76 (0.92) Diff: – Eq: $\dagger$	0.229193633	0.030 (0.651)

systems received high scores (4.31 for SYS1 and 4.01 for SYS2). This confirms that they successfully supported users in managing activities.

As discussed in Section 6.4, when prompted to provide open-ended feedback, the theme *Time pressure and increased productivity* emerged more prominently in SYS0 than in the other systems (see Section 6.8 for the complete analysis of themes). Conversely, users reported increased productivity with SYS1 and SYS2.

Overall, these results support HP3.

6.5. HP4: Enhancing the autonomy of agentic AI for productivity service integration negatively impacts user trust in the system.

To address HP4, we examined user trust through the Trust in Automation (TiA) post-task questionnaire. Following Körber (2019)'s indications, we analyze results in an aggregated form through the dimensions reported in Table 10 and Fig. B.3 (the latter is in the Appendix). We assessed trust across the different dimensions of the questionnaire.

Results reveal no statistically significant differences across the three systems in any trust dimension. Notably, statistical equivalence testing shows that trust is consistent across systems regardless of their autonomy levels:

- For Reliability/Competence, both AI-driven systems (SYS1: $M=3.50$, $SD=0.66$; SYS2: $M=3.52$, $SD=0.71$) have higher mean scores than SYS0 ($M=3.26$, $SD=0.71$), with statistical equivalence between SYS1 and SYS2.
- Understanding/Predictability is statistically equivalent across the three systems. The scores slightly decrease as automation increases (SYS0: $M=2.81$, $SD=0.53$; SYS1: $M=2.69$, $SD=0.57$; SYS2: $M=2.63$, $SD=0.49$).
- The Familiarity construct does not have any statistically significant differences or equivalences. The scores decline slightly as automation increases. This is expected because SYS0 is the most familiar system, being inspired by traditional productivity services. Moreover, the user interface is fairly consistent across systems. Thus, the effect of AI intervention on familiarity is likely mitigated.
- For the Intention of Developers, the AI-driven systems received higher ratings (SYS1: $M=3.60$, $SD=0.70$; SYS2: $M=3.59$, $SD=0.81$) than the baseline (SYS0: $M=3.47$, $SD=0.72$), with statistical equivalence between pairs of systems.
- The Trust in Automation dimension shows the most promising results for the AI-driven systems. Mean scores are higher for SYS1 ($M=3.75$, $SD=0.83$) and SYS2 ($M=3.76$, $SD=0.92$) than SYS0 ($M=3.39$, $SD=0.97$). There is statistical equivalence between SYS1 and SYS2.

These findings reveal that users' trust in the systems did not decrease significantly, despite AI-driven automation and increasing levels of system autonomy. Indeed, we observed numerical increases in trust.

Thus, **HP4 is not supported**: users maintained equivalent levels of trust in SYS0, SYS1, and SYS2 regardless of the increasing autonomy of the AI-driven systems.

6.6. HP5: The trust in agentic AI for integrating productivity services with different levels of autonomy negatively correlates with the desirability of control.

To address HP5, we conducted regression analyses between the results of the Desirability of Control questionnaire, administered before participants interacted with the systems, and the Trust in Automation (TiA) constructs measured after the interaction. Table 11 presents the correlation coefficients across all systems and trust dimensions. We consider the following thresholds for correlation magnitudes: $|r| < 0.2$ negligible, $0.2 \leq |r| < 0.4$ weak, $0.4 \leq |r| < 0.6$ moderate, and $|r| \geq 0.6$ strong (Akoglu, 2018).

Table 11

Correlations between desirability of control (Burger & Cooper, 1979) (administered before interacting with systems) and Trust in Automation (TiA) constructs (measured after interaction) for SYS0, SYS1, and SYS2.

	SYS0	SYS1	SYS2
Reliability/Competence	0.41267	0.29602	0.17499
Understanding/Predictability	−0.28845	−0.04211	−0.19734
Familiarity	0.01402	0.07812	0.04457
Intention of Developers	0.22722	0.25896	0.23036
Trust in Automation	0.35162	0.24513	0.18565

- Reliability/Competence measures users' perception of the system's ability to interpret situations correctly, work reliably, and perform tasks. For this dimension, we observed a moderate positive correlation in SYS0 ($r = 0.41267$) that progressively weakened in SYS1 ($r = 0.29602$) and SYS2 ($r = 0.17499$). The decreasing correlation across systems with increasing autonomy suggests that the relationship between the Desirability of Control and perceived system reliability/competence becomes less pronounced as automation increases. In other words, users' desire for control becomes less predictive of how they evaluate system reliability as the system is more autonomous. Notice that, in Table 10, SYS1 and SYS2 achieved higher mean scores for this dimension (3.50 and 3.52, respectively) compared to SYS0 (3.26). This suggests that regardless of participants' differences in Desirability of Control, our AI-driven systems were perceived as more reliable than SYS0.
- Understanding/Predictability was the only dimension showing consistent negative correlations with Desirability of Control across all systems, partially aligning with HP5. However, these correlations (all negligible or weak) varied in strength (SYS0: $r = -0.28845$, SYS1: $r = -0.04211$, SYS2: $r = -0.19734$) and did not follow a clear linear pattern with increasing system autonomy. The lack of correlation for SYS1 indicates that combining automated action suggestions and user confirmation may be a good compromise between increased productivity and user control. This aligns with qualitative feedback (Section 6.8, theme *Automation and User Control*), where participants praised SYS1's balance of automation with user control. In other words, a system that suggests automated actions while supporting user intervention appears to satisfy users regardless of their Desirability of Control. Notice that the theme *Automation and User Control* was also prevalent for SYS2. Nevertheless, some participants complained about the lack of control when interacting with that system.
- For the Intention of Developers, we observed weak positive correlations across all systems (SYS0: $r = 0.22722$, SYS1: $r = 0.25896$, SYS2: $r = 0.23036$). We remind the reader that the Intention of Developers measures trust in the system creators' motivations and ethics. Our findings indicate that users' perception of developers' intentions was weakly related to their Desirability of Control, regardless of system autonomy.
- The Trust in Automation dimension was correlated positively with the Desirability of Control. However, the correlation became weaker with increasing system autonomy (SYS0: $r = 0.35162$, SYS1: $r = 0.24513$, SYS2: $r = 0.18565$). Trust in Automation directly measures users' explicit trust statements and willingness to rely on the system. Similar to the relationship between Desirability of Control and Reliability/Competence, the weakening correlation as autonomy increases suggests that the relationship between Desirability of Control and overall trust becomes less pronounced in more autonomous systems. Moreover, looking at Table 10, Trust in Automation is higher for SYS1 and SYS2 than SYS0. This suggests that the first two systems induced more homogeneous and higher scores than the third one in this subscale.
- The Familiarity dimension was not correlated with the Desirability of Control.

HP5 is not supported. Contrary to our hypothesis, most dimensions showed positive correlations with the Desirability of Control. However, the consistent pattern of decreasing correlation strength as system autonomy increases suggests that the relationship between Desirability of Control and trust changes as AI-driven automation becomes more prominent in the systems.

6.7. HP6: The trust in agentic AI for integrating productivity services with different levels of autonomy correlates with propensity to trust systems.

We conducted regression analyses between the Propensity to Trust construct of the TiA questionnaire, administered before participants interacted with the systems, and the Trust in Automation (TiA) constructs measured after the interaction. Table 12 shows the results of this analysis, considering the same thresholds as in Section 6.6.

- For the Reliability/Competence dimension, we observed a U-shaped pattern: the (positive) correlation was weak to moderate in SYS0 ($r = 0.37952$), substantially lower in SYS1 ($r = 0.08001$), and moderate in SYS2 ($r = 0.4881$). As reported in Table 10, the standard deviation for this subscale is the lowest in SYS1, revealing more homogeneous scores regardless of participants' Propensity to Trust.

The correlation pattern suggests that, as SYS1 presents suggestions that the user can manage, it allows people to assess system capabilities, reducing the influence of their intrinsic trust tendencies. In contrast, when operations are manual (SYS0) or the system performs actions autonomously (SYS2), users depend more heavily on their trust dispositions in evaluating system reliability. It is worth noting that SYS2 automatized low-stakes actions depending on the user's confirmation of the email to be sent. However, some participants felt that, even for this type of action, they had to take care that the system did not fail.

Table 12

Correlations for each system between Propensity to Trust (construct of the TiA questionnaire) and the other post-task TiA constructs.

	SYS0	SYS1	SYS2
Reliability/Competence	0.37952	0.08001	0.48810
Understanding/Predictability	−0.01938	0.14311	−0.05837
Familiarity	−0.11128	0.09316	0.03862
Intention of Developers	0.43743	0.27781	0.45251
Trust in Automation	0.44820	0.20258	0.51892

We may hypothesize that SYS1 was perceived as more reliable and competent than SYS0 because it exhibited its ability to interpret the current context and suggest the best course of action while leaving the user in control of operations. This was probably exacerbated by the time pressure we imposed on participants during the test. See [Section 6.8](#), theme *Time pressure and increased productivity*, and question N3 in [Table 8](#).

- The Understanding/Predictability dimension showed negligible correlations across all systems. This indicates that Propensity to Trust was poorly related to participants' perception of the system's predictability, regardless of their level of autonomy.
- For the Intention of Developers, we observed another U-shaped pattern with moderate positive correlations in SYS0 ($r = 0.43743$) and SYS2 ($r = 0.45251$), and a slightly weaker correlation in SYS1 ($r = 0.27781$).

The strong correlations concerning SYS0 and SYS2 suggest that when a system is either manual (SYS0) or highly autonomous (SYS2), users rely strongly on their trust dispositions to evaluate developers' intentions. In contrast, the user can check all the actions suggested by SYS1. This brings transparency and control; thus, users can judge developers' intentions based on direct evidence rather than on their trust tendencies.

- The Trust in Automation dimension displayed a pronounced U-shaped correlation pattern. A moderate positive correlation in SYS0 ($r = 0.4482$) decreased in SYS1 ($r = 0.20258$), then highly increased in SYS2 ($r = 0.51892$).

Trust in Automation measures users' explicit trust statements and willingness to rely on the system. The correlation pattern reveals that SYS2 created the strongest link between participants' trust tendencies and resulting trust in the system. This confirms that, in highly autonomous systems, as SYS2, trust disposition plays a decisive role in system acceptance. In contrast, the significantly lower correlation in SYS1 shows that explicitly leaving users in control of the system's actions through confirmation or manipulation creates a trust-building mechanism that works effectively regardless of their trust dispositions. This makes SYS1 more inclusive than SYS2 for people with varying Propensity to Trust.

The higher correlation with SYS0 rather than SYS1 may seem counterintuitive. However, it can be explained by the support that SYS1 provides. Based on [Hegner et al. \(2019\)](#), its usefulness might have helped foster user trust, that is, SYS1 might have been considered as intelligent without compromising user control. Consistently, in the results of [Table 10](#), SYS1 has the lowest standard deviation for the Trust in Automation subscale. This suggests SYS1 led to higher and more homogeneous Trust in Automation scores, regardless of participants' Propensity to Trust. Moreover, the mean scores for SYS1 and SYS2 are very similar and higher than for SYS0.

These findings align with the open feedback themes. Participants interacting with SYS0 expressed more frustration with time pressure (*Time pressure and increased productivity* theme). Differently, those using SYS1 highlighted the benefits of automation with control (*Automation and User Control*). See [Section 6.8](#) for details.

- The Familiarity dimension exhibited no correlations with Propensity to Trust.

These findings support HP6, confirming that Propensity to Trust correlates with post-task trust measures across systems. However, the strongest correlations concern the highly automated system (SYS2). The lower correlations for SYS1 imply more homogeneous trust scores, regardless of users' Propensity to Trust.

6.8. Analysis of open feedback

We identified inductively the main themes that emerge from participants' open-ended feedback after interacting with the systems through Thematic Analysis ([Braun & Clarke, 2006](#)). This analysis integrates the results of the post-task questionnaires with aspects emerging from participants' spontaneous descriptions.

Three authors of this paper independently annotated the open-ended feedback to identify recurring themes. To ensure inter-coder reliability, they compared their coding results and discussed to resolve discrepancies until they reached consensus on themes. In detail, the researchers first read and discussed the free-text answers to acquire familiarity with the data. Then, they organized the answers systematically (depending on the topics) and selected each sentence that provided feedback on the interaction with the systems. After that, they grouped the selected sentences into themes and discussed whether these themes made sense, revising them until they reached an agreement.

Following this, they independently annotated each comment based on the agreed-upon themes, and reached a Cohen's Kappa of 0.703. Then, a consensus was established through communal discussion. In the following, we discuss the themes identified. For each theme, the counts we report are based on the number of participants who mentioned it.

Ease of use. This theme was prominent, particularly in SYS0, with 17 positive comments praising its intuitive user interface. For example, P408 declared, “The interface was very user friendly. Everything was clear and easy to understand.”. These comments highlight that SYS0 was well-designed, and functionalities were intuitive. This is important because the other systems are built on top of this one.

The user-friendly design was also appreciated in SYS1, with 9 positive comments. Although this was a new system for the participants, it was not particularly difficult to use. P595 said “Honestly i think the interface looks nice and easy to use actually its my first time using something like this and i must say i would love to have use it”. The same is for SYS2. In this case, the theme was mentioned 8 times, and P182 said: “The task was simple and easy to do. The use of auto schedule and respond made it very easy”. This feedback shows that user-friendliness was successfully maintained across all systems, facilitating a smooth user experience even as additional features were introduced.

Familiarity. Five participants mentioned that SYS0 was familiar. For example, P507 said “That is functioned like other modern interfaces (google calendar, etc). Simple. Fast.” Nobody mentioned this theme after interacting with the other two systems. This suggests that they were perceived as novel.

Time pressure and increased productivity. The frequency of this theme was 11 occurrences for SYS0, 7 for SYS1, and 3 for SYS2. Specifically, P609 used SYS0 and complained, “It was kinda rushed”. Differently, increased productivity was mentioned 9 times with SYS2, 5 times with SYS1, and never with SYS0.

Automation and User Control. The automation of productive services (draft email generation, calendar, and to-do management) occurred 40 times for SYS1 and 45 for SYS2 (0 times for SYS0, which lacks automation), with positive evaluations. P598 said about SYS1, “The system was fast in responding (creating responses), creating events and managing my schedule. It saves a lot of time for people with busy schedules and limited time.” Participants appreciated the automatic email drafting feature. For example, P514 declared, “The ability to create draft emails on command was the most useful, it sounded professional too.” Moreover, they liked the automatic management of the calendar and to-dos, as mentioned by user P136, “All of it! The AI assistant, to do and calendar assistance was much appreciated!”. Similarly, participants appreciated SYS2. For example, P86 stated, “I really liked that all the meetings and tasks were automatically forwarded to the calendar and to-do list. I also liked that the interface was able to recognise some events that were planned the same time to which another was proposed.” P437 said: “I like the fact that it automatically updated my to do list and my calendar as that helped me respond to the emails quickly. The idea of having a system that is capable of generating automatic yet original responses is amazing”.

The theme of automation is related to user control. The latter was mentioned 9 times for SYS2, and never for the other two systems. We expected that some participants would have asked for more control over a highly autonomous system, regardless of the transparency support provided by the notifications. For example, P137 said, “Verifying that no mistakes are made. If the system even makes a very few mistakes, I need to validate everything to ensure no mistakes happen.”

Conflict management. Participants discussed the proactive conflict management capabilities of SYS1 and SYS2: 3 and 9 people commented on this feature, respectively. The system’s ability to solve scheduling conflicts was appreciated. For instance, P515 commented on SYS2: “I liked how it tried to work around my busy schedule, especially how it helped me reply faster.” Moreover, some participants suggested improvements to the schedule management to prevent user overload. For instance, P124 proposed: “Suggesting different meeting times when your calendar is getting quite busy (not that you have a conflict, but that too many meetings in a day would leave you unable to do your regular work).” Similarly, P458 recommended: “Some way to have a limit on amount of tasks that could be taken in one day and a way to decide which ones are mandatory and which ones could be rescheduled on to another day.” These suggestions highlight that when scheduling user commitments, we should consider the user’s overall workload, the time needed to accomplish the requested work, and task prioritization. This is a complex function and is part of our future work.

Email tone. This theme concerns the email tone in email draft generation. It gained some attention in SYS1 (2 times) and SYS2 (3 times). Participants expressed interest in receiving multiple draft options with different tones. For example, P484 stated, “it was fine, maybe there could be several options of the email, so that I could select the one that I prefer, they could be written in different ways, more formal or informal for example”. Moreover, some users noted issues with the writing style of AI-generated content. Specifically, P559 complained, “The responses were a little verbose, a little too ‘AI’”. P128 declared, “The mails themselves are a bit robotic and dry”. This feedback highlights the ongoing challenge for LLMs to produce more natural, human-like text. It is worth noting that these stylistic deficiencies might be addressed through specialized prompt engineering techniques, for instance, by asking for shorter descriptions or more enthusiastic ones. Strategic refinement of input prompts may steer LLMs to produce text with more natural linguistic patterns and more concision.

6.9. Discussion of results

This section discusses the results of the user test, considering our research questions and hypotheses.

- RQ1: *Does the autonomy level of an Agentic AI system improve user activity when managing multiple productivity services?*
The corresponding hypothesis *HP1* is supported by the results. Thus, we answer this research question positively.

Performance metrics reveal that both AI-driven systems (SYS1 and SYS2) outperformed the baseline (SYS0) in user activity, message response rates, and accuracy. SYS2 achieved the highest throughput with an average of 24.92 actions during the simulation, compared to 18.22 for SYS1 and 6.60 for SYS0. Moreover, the observed differences among systems in message response rate were consistent with these results. The accuracy of task execution also improved with AI assistance, as evidenced by the higher precision and significantly higher recall scores of SYS1 and SYS2, compared to SYS0.

Overall, our findings suggest that AI-automated systems sensibly improve accuracy compared to the manual baseline (SYS0), while significantly increasing throughput.

These results confirm that previous findings on the usage of AI (Qian & Wexler, 2024) in the Software Engineering domain extend to the integration of productivity applications.

- RQ2: *Does the autonomy level of an Agentic AI system reduce user workload when managing multiple productivity services?*

Our results confirm HP2, and we answer this research question positively.

The workload assessment based on the NASA-TLX questionnaire reveals a significantly reduced cognitive burden during interaction with our AI-driven systems. Participants reported lower mental and temporal demands, effort, and frustration when using SYS1 or SYS2 than with SYS0. Moreover, the aggregate NASA-TLX score decreases from SYS0 to SYS2. Furthermore, by monitoring participants' actions on the user interfaces of SYS1 and SYS2, we found that they accepted almost all the counterproposals generated by the systems to solve calendar conflicts.

As above, these results extend previous findings (Qian & Wexler, 2024) to the integration of routine productivity applications.

- RQ3: *Does the autonomy level of an Agentic AI system increase user perceived support when managing multiple productivity services?*

Our results support HP3. Thus, we answer this research question positively.

The results of the post-test questionnaire on user perceived support confirm that both SYS1 and SYS2 provide a high, mostly equivalent, perceived support. Moreover, considering participants' open feedback, results suggest that SYS0 induced a lower perception of support.

This extends previous work in the Software Engineering domain (e.g., Wyrich et al. (2020) on the use of refactoring bots) to productivity applications in multitasking environments.

- RQ4: *Do different levels of AI autonomy influence user trust in the system?*

We answer this research question negatively because HP4 was not supported.

Indeed, users maintained equivalent levels of trust regardless of the increasing autonomy of systems. The Trust in Automation (TiA) measurements show stable or slightly improved trust levels across all dimensions as automation increases. We found statistical equivalence between systems in most trust dimensions. In particular, SYS1 and SYS2 consistently received higher numerical ratings than SYS0 in Reliability/Competence, Intention of Developers, and overall Trust in Automation. Since in our experiment participants were familiar with the domain of the tasks to be performed, this finding challenges the conventional assumption that increasing automation can erode expert users' trust in the AI (Qian & Wexler, 2024).

- RQ5: *Does Desirability of Control moderate the impact of different levels of AI autonomy on user trust in the system?*

HP5 was not supported.

Our analysis reveals a positive correlation between most dimensions of participants' Desirability of Control and the autonomy of our systems. Desirability of Control correlates positively with Reliability/Competence and Trust in Automation. However, the correlation decreases as autonomy increases. This suggests that the relationship between Desirability of Control and overall trust becomes less pronounced as autonomy of operation increases.

Qualitative feedback collected from open-ended questions is consistent with these findings. The main emerging themes confirm that participants expressed positive evaluations of the automation and proactive conflict detection functions offered by SYS1 and SYS2. However, opinions are more varied in SYS2. In this case, some users expressed a need to verify system behavior while others appreciated the combination of autonomy and transparency, which enables users to compensate for undesired behaviors. This suggests that, as AI autonomy increases, coupling autonomy with explanation (even a basic one, as we introduced in our systems) is key to acceptance of technology.

- RQ6: *Does Propensity to Trust systems moderate the impact of different levels of AI autonomy on user trust in the system?*

HP6 was supported by the results. Thus, we answer this question positively.

Our findings confirm that the Propensity to Trust is positively correlated with the other trust dimensions of TiA. As discussed, post-task questionnaire results reveal that this personality trait was the least influential in SYS1. This suggests that SYS1 might foster more trust than the other systems. We believe that the system's ability to generate actionable elements, combined with the complete user control it guarantees, contributes to this effect.

The open-ended feedback provided by participants reinforces these findings by (i) highlighting the ease of use across all systems and (ii) praising the time-saving aspects of automation in SYS1 and SYS2. The theme of increased productivity was often mentioned in the AI-driven systems. Moreover, concerns about time pressure decreased as automation increased. Some users of SYS2 expressed a desire for greater control, suggesting a potential optimization point between the benefits of automation and user agency. These findings reveal that carefully designed Agentic AI can reduce workload and improve task performance while maintaining user trust.

It is worth mentioning that, even though the experiment was conducted in a simulated environment, it enabled participants to achieve immersion in the experimental setting. The high number of actions they performed (averaging from 6.60 to 24.92 actions across systems during the 5-min simulation) and the quality of their email responses demonstrate meaningful engagement with the tasks. Our manual analysis also revealed that participants responded to emails coherently, staying on topic with incoming message

content in 94.8% up to 97.3% of cases across all systems. This indicates that they successfully interpreted and responded to the work scenarios as intended. The diversity of professional backgrounds represented in our participant pool (spanning 10 job categories) and the domain-specific email scenarios ensured that participants encountered familiar terminology and realistic workplace situations, facilitating natural task engagement despite the simulated environment.

7. Implications

In this section, building on the results, we outline the theoretical and practical implications, and the design principles.

7.1. Theoretical implications

The results of our study yield implications for designing and exploiting Agentic AI systems in productivity environments. We describe the theoretical perspectives that emerged from the results.

- *Automation acceptance.* The successful integration of increasing levels of AI agency without eroding user trust challenges existing assumptions about automation acceptance, like the one by [Qian and Wexler \(2024\)](#) concerning expert users' trust in AI. We found that user control support is a key element to enforce trust in Agentic AI systems with increasing levels of autonomy. This suggests that theoretical models of technology acceptance may need refinement to account for the unique characteristics of modern Agentic AI systems that prioritize user control and transparent operation.
- *Negotiated agency.* Our findings confirm how the traditional boundaries between human and machine decision-making become increasingly permeable in Agentic AI systems. The observed user acceptance of delegation in our systems suggests that technology is progressively viewed less as a tool and more as a collaborator with shared agency ([Lindgren, 2025](#)). We found a similarity in user trust between medium automation with explicit confirmation (SYS1) and high automation, where low-stakes tasks are performed autonomously but the user must confirm high-stakes ones (SYS2). This indicates that designers have flexibility in determining the appropriate level of automation based on context. For example, when sending emails, explicit confirmation may be preferable even though it requires additional user actions. Our results suggest a "negotiated agency" model where, similar to [Horvitz \(1999\)](#), control is dynamically allocated based on the type of actions to be performed. However, we introduce user characteristics related to the Desirability of Control and the Propensity to Trust systems to refine the strategy to allocate control. This model aligns with the principle of Human-Centered AI ([Shneiderman, 2022](#)).
- *Appropriate reliance.* The concept of "appropriate reliance" ([Lee & See, 2004](#)) gains renewed relevance in our findings. Their framework argues that trust guides, but does not completely determine, reliance on automation. Our findings corroborate this by demonstrating how appropriate reliance can be maintained even as system autonomy increases, provided that the design supports what Lee and See term "calibration, resolution and specificity of trust". Specifically, our experiments demonstrated nuanced calibration of trust relative to system autonomy, contrary to concerns that an automation bias leads to uncritical acceptance of AI suggestions ([Hao-Ping et al., 2025](#)). Indeed, users discerned in their delegation patterns, as supported by the correlation patterns between Desirability of Control and Trust in Automation. This confirms the findings of [Brachman et al. \(2025\)](#) that theoretical frameworks of human-AI collaboration must account for users' mental models of AI capabilities and limitations.
- *User interface design.* Our qualitative findings reveal that the "Ease of use" theme emerged prominently across our systems, with participants consistently highlighting the intuitive nature of their user interfaces. This enforces the importance of maintaining familiar user interfaces (as we did in our systems) to facilitate user adoption of advanced automation capabilities ([Davis & Granic, 2024](#)).
- *Situation awareness.* Systems exhibiting situation awareness (e.g., by detecting calendar scheduling conflicts and advising the user accordingly) can improve user confidence. This suggests that an investment in robust context modeling capabilities may enhance user acceptance.

7.2. Practical implications

The significant improvements in throughput and accuracy offered by Agentic AI systems have important implications for organizational productivity. By offloading routine cognitive tasks to AI systems, knowledge workers might redirect their attention to higher-value activities that require human judgment and creativity. This represents a fundamental shift in the organization of knowledge work. The improved accuracy in user activity observed in our AI-driven systems suggests potential to reduce costly errors in professional communications and scheduling. This could have cascading benefits for organizational efficiency.

However, delegating communication and scheduling decisions to AI systems raises important ethical considerations. As the autonomy of AI systems increases in professional communications, questions arise about accountability for decisions and representations made on behalf of users ([Busuioc, 2021](#)). Clear attribution and review mechanisms may be necessary. Moreover, the context integration that makes these systems effective involves processing possibly sensitive communications and scheduling information, highlighting the importance of robust privacy protections and data governance frameworks.

These implications suggest broader trajectories for how Agentic AI might transform productivity environments while respecting user autonomy and maintaining appropriate human oversight. The clear progression in efficiency and accuracy as the AI agency increases suggests that well-implemented AI-driven systems can mitigate the tension between speed and accuracy in productivity applications, allowing for improvements in both dimensions. This potential to address challenges in maintaining shared context

across productivity applications without imposing additional cognitive burdens on users represents a significant advancement in human-centered AI design.

Our findings also suggest a significant transformation of the workforce. As Agentic AI systems become more capable of handling routine tasks, organizations will need to develop new strategies for workforce development that emphasize human capabilities, such as creative problem-solving, ethical reasoning, and managing interpersonal relationships. Thus, integrating these systems may require organizational restructuring to optimize workflows that combine human and AI contributions. However, this requires change management strategies, emphasizing transparent communication about system capabilities and limitations, and does not exclude the need for organizational support in terms of training and resource management. For example, Wang and Zhang (2025) found that AI-driven innovative behavior mediates the relationship between technological/organizational factors and employee outcomes in hospitality management.

Our findings could extend beyond routine productivity apps to domains where generative and Agentic AI is becoming pervasive. For example, in education, partially autonomous AI systems that suggest actions while preserving user confirmation can support tasks such as feedback drafting and coursework management, aligning with prior work on learner control and human-centered AI (Brusilovsky, 2023). In healthcare administration, Agentic AI could reduce clinicians' workload by assisting with low-stakes coordination tasks such as appointment scheduling and follow-ups (Jo et al., 2023). However it should maintain human control on high-stakes activities. In hospitality management, management Generative AI can improve efficiency and guest experiences by enhancing the creativity of employees in marketing campaigns and designing solutions (Wang & Zhang, 2025).

Finally, our results highlight accessibility implications. The reduced cognitive load offered by Agentic AI systems may benefit neurodivergent individuals or those with cognitive processing challenges, creating more inclusive work environments. Organizations should consider leveraging these technologies as workplace accommodation tools that expand employment opportunities for diverse populations.

7.3. Design principles

Building on these results, on prior work on mixed-initiative interaction (Horvitz, 1999) and human-centered AI (Shneiderman, 2022), we outline the following design principles:

- Use “suggest-then-confirm” patterns that leverage the use of draft actions and an explicit confirmation, particularly for non-expert users and for high-stakes tasks. This aims to support appropriate reliance in Agentic AI (Hoff & Bashir, 2015; Lee & See, 2004) and progressive control handover (Parasuraman et al., 2000).
- System autonomy should be treated as an adjustable feature. We suggest beginning with confirmable suggestions, then allowing users to opt into higher autonomy as they gain confidence. Mixed-initiative strategies support this incremental delegation while preserving user agency (Horvitz, 1999).
- Provide awareness of what the system proposes and does, as well as of the outcome of the actions performed. This helps users to calibrate reliance (Brachman et al., 2025; Kim et al., 2024) in Agentic AI. Because explanations can be misleading when systems are unreliable, explanation mechanisms should be conservative (Cau et al., 2023; Sadeghi et al., 2024).
- Since trust formation depends on individual differences, systems should allow autonomy features (e.g., confirmation frequency, proactive execution thresholds, explanation verbosity) to be configured explicitly and adapted over time while preserving user control (Brusilovsky, 2023).

8. Limitations

Despite the promising results of our study, we acknowledge some limitations.

1. While our AI-driven systems achieved high accuracy in processing emails and managing tasks, LLMs can occasionally generate credible but incorrect output. This raises concerns about the eventuality of misplaced user trust, especially in the highly automated system (SYS2), where users might not thoroughly verify automated actions before execution. As noted by Chan et al. (2023), LLMs can present fabricated information with high confidence, potentially leading users to accept incorrect suggestions without double-checking. Although the factuality of LLMs tends to improve with their size (Tam et al., 2023), hallucinations present risks in professional environments where communication errors could have substantial consequences. These issues may be more or less prevalent in different contexts, such as language and culture (Shen et al., 2024), suggesting the need for validation mechanisms in AI-driven productivity systems. Nonetheless, it is important to note that our proposed framework is agnostic to the specific LLM model used. As these models improve, we expect they will mitigate this issue.
2. Our experiment captured immediate user reactions and short-term performance effects within a limited time interval. Thus, the long-term effects of interacting with Agentic AI systems for productivity management remain unexplored. Extended use might reveal different patterns of trust formation, skill degradation, or overreliance (Hao-Ping et al., 2025) that were not observable in our experimental sessions. Prolonged reliance on AI assistance could diminish users' problem-solving abilities in certain contexts. Moreover, the observed increase in task performance with higher autonomy levels, while maintaining user trust, could be problematic in high-stakes environments. Longitudinal studies tracking changes in user behavior, trust dynamics, and skill retention over weeks or months would enhance our understanding of the impact that AI-driven systems have on professional practices.
3. Our empirical results should be interpreted within the specific context of our study design. Specifically, the outcomes we observed reflect the intersection of appropriate task complexity and appropriate stakes. The combination of short interaction time and

simulated scenarios could have minimized the visibility of certain risks (e.g., skill degradation, overreliance, cascading errors from LLM hallucinations). Future research should systematically vary these contextual factors, such as task complexity, stakes, time pressure, and interaction duration, to map the boundaries within which Agentic AI systems like ours deliver benefits without incurring risks.

4. While our simulated scenarios were designed to mirror authentic workplace communications and participants demonstrated high engagement through coherent responses and substantial task completion, the controlled experimental environment may not fully capture the complexities of real organizational contexts. Indeed, our participants interacted with pre-designed scenarios rather than managing their real work communications, which might have influenced their perception of system performance and risks. We mitigated this by providing scenarios as close as possible to each participant's field of work. A field study conducted in authentic work environments would complement our controlled experiment by capturing how users integrate these systems into their established workflows and organizational processes. Moreover, it might reveal user preferences for the level of systems' autonomy and the tone of AI-generated content suitable for the communication stake and the social relationships between peers.
5. We designed the simulated scenarios to resemble genuine workplace communications, and participants from diverse backgrounds engaged thoroughly with the tasks. However, the controlled experimental environment did not fully account for the complexities inherent in real organizational contexts. A more granular approach is needed to investigate how distinct professional roles and hierarchies influence user interactions with Agentic AI. Professional differences might also imply some diversity in participants' prior knowledge of productivity applications and AI. In a survey, [Wang and Sun \(2025\)](#) found that previous expertise impacts university students' acceptance of intelligent personal assistants in education. Thus, it is worth investigating the role of this variable to extend our results on the relationship between system autonomy and user trust.
6. Although our participant sample was geographically diverse, this study did not analyze cultural background as moderating variables. Because the present study was not designed or powered for subgroup analyses, the influence of these characteristics on the autonomy of Agentic AI will be studied in future work.
7. The Conflict Finder we developed is a reliable tool to reveal conflicts in the user's calendar. However, the scheduling of activities should consider the overall set of existing commitments to avoid overloading users with too many events and deadlines in a single day. As mentioned in [Section 6.8](#), further research is needed to schedule activities by evaluating the user's overall workload, the time to accomplish commitments, and task prioritization.

9. Conclusions and future work

This study investigated the impact of varying levels of AI agency on user activity, workload, perceived support and trust in the integration of productivity applications. Our findings reveal that Agentic AI assistants can significantly enhance productivity while maintaining or improving user trust, depending on individual characteristics concerning the Desirability of control and the Propensity to Trust.

For our experiments, we developed three systems that integrate productivity services and have different autonomy levels. The Agentic AI with medium (SYS1) and high (SYS2) autonomy outperformed the baseline system (SYS0) across key performance metrics while reducing user workload. Both AI-driven systems achieved equivalent or slightly higher user trust than the baseline. Moreover, we found that systems that balance automation with user control (like SYS1) can accommodate the widest range of individual user characteristics related to Desirability of Control and Propensity to Trust. Participants reported increased productivity, reduced time pressure, and appreciation for the conflict detection functions offered by the two AI-driven systems. These results have important implications for the design of AI-driven systems that integrate productivity services. Designers should focus on transparent operation, contextual awareness, and appropriate confirmation mechanisms. Our findings suggest that Agentic AI can address the traditional tension between efficiency and user control, offering a promising path forward for integrating AI assistance into professional workflows. Several promising directions for future research emerge from our findings:

- Prior work shows that trust in automation develops through experience and may change as users encounter system errors or limitations ([Lee & See, 2004](#)). Users may refine their mental models of system functionalities or adjust their reliance on the system. Longitudinal studies would enhance our understanding of how user trust evolves over extended periods of interaction with Agentic AI systems.
- Expanding the range of productivity applications integrated through Agentic AI would provide a more comprehensive understanding of the approach's potential. While our study focused on email, calendar, and to-do list management, future research could incorporate additional tools such as document editing and project management systems. Our proposed framework is designed to integrate services seamlessly, and this expanded scope would better reflect the complexity of modern work environments. Additionally, providing more services would lead to collecting valuable data on user interaction with AI-driven systems.
- Replicating our study across culturally distinct populations, different professional roles, and organizational environments, as well as conducting field studies in real workplaces, would improve the generalizability of the present findings. For example, in individualistic cultures, users tend to seek information from reputable sources when evaluating an innovative technology. In collectivistic ones, they are more likely to rely on subjective assessments ([Lee et al., 2013](#)).
- We recognize the importance of broadening the scope of individual user characteristics examined to connect different levels of system autonomy and user trust. While the present study focused on psychological constructs such as Desirability of Control and Propensity to Trust, future research should also investigate how demographic variables (including gender, age, and geographic location), technological proficiency, prior AI experience and cultural background may moderate the relationship between AI

autonomy. Expanding the range of user characteristics in future studies will enhance the generalizability of our findings. Future research could also explore adaptive systems that adjust their autonomy based on detected user characteristics or explicit user configuration.

- While our study focuses on trust as the main variable, trust could be related to users' behavioral intentions and reliance on automated systems. Future studies should explicitly include action intention variables, such as users' intention to let the system autonomously execute a suggested action. This would also support deeper investigation of appropriate reliance and overreliance dynamics across varying task stakes and autonomy levels.
- As mentioned in Section 2, some recent work investigates tone-aware content generation based on LLMs but focuses on explaining recommender system results (Okoso et al., 2025). Extending that type of work to email generation could be an interesting path for our research.
- We focused on providing a concise interaction model because in the context of productivity, time pressure and multitasking environments could make longer explanations counterproductive. More elaborate explanations might have influenced trust dynamics differently, particularly for SYS2, where high autonomy increases stakes. For instance, explanations of *why* the system selected specific calendar slots (e.g., "2 PM chosen because it provides a 30-min buffer before your next meeting") might enhance users' mental models of system capabilities. Future research should explore different explanation levels to identify optimal designs across different autonomy levels, task stakes, and user characteristics.
- Finally, a deeper investigation about the stakes and types of tasks should be carried out. For example, routine low-stakes tasks may be well served by systems with higher levels of autonomy. In contrast, creative or high-stakes tasks are more likely to require explicit user confirmation and more user control. Moreover, in collaborative settings, where errors can affect multiple stakeholders, system autonomy should be treated carefully.

CRedit authorship contribution statement

Angelo Geninatti Cossatin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Fabio Ferrero:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Liliana Ardissono:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Noemi Mauro:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Data availability

The data is publicly available in a repository which is linked in the paper

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Perplexity¹¹, DeepL¹², and Grammarly¹³ for the execution of the following tasks: Paraphrase and reword, Text Translation, and Grammar and spelling check. After using these tools, the authors reviewed and edited the content as needed. The authors take complete responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Prolific instructions

Below, we report the instructions given to Prolific workers before accepting to partake in the study:

This project aims to analyze the impact on users of the support offered by LLM-powered agents in professional environments. To participate in the study, we require that you have experience with routine productivity applications in the workplace. This study may contribute to the areas of Artificial Intelligence, Large Language Models, Information Visualization and Recommender Systems.

Please notice that the objective of the experiment is to identify possible faults in the proposed service, and not to test your own ability, or intelligence.

¹¹ <https://www.perplexity.ai/>

¹² <https://www.deepl.com>

¹³ <https://app.grammarly.com/>

Please note that you cannot leave the page once the experiment starts, until you submit your work to Prolific. If you leave the page you will receive multiple warnings. Failure to comply with this even after the warnings will result in automatic disqualification. Similarly, prolonged inactivity during the experiment is not allowed and will result in automatic disqualification after multiple warnings. Thus, you should start the experiment when you have the time to complete it (Tables C.1–C.3).

Appendix B. Graphical representation of evaluation results

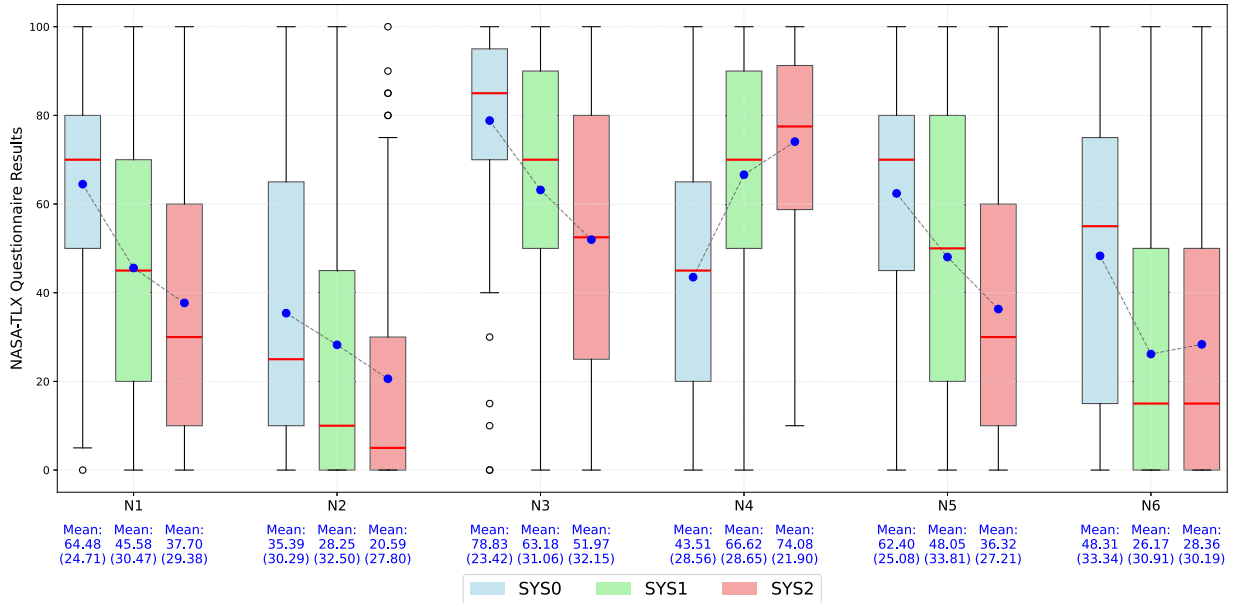


Fig. B.1. NASA-TLX workload assessment results across dimensions N1-N6 of Table 8 for systems SYS0, SYS1, and SYS2. Box plots show the distribution of ratings for each system. Red lines indicate medians, and blue dots mark means. Dashed lines connect mean values across systems to highlight performance trends. The empty circles denote the outliers. The mean values and standard deviations are displayed below each box plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

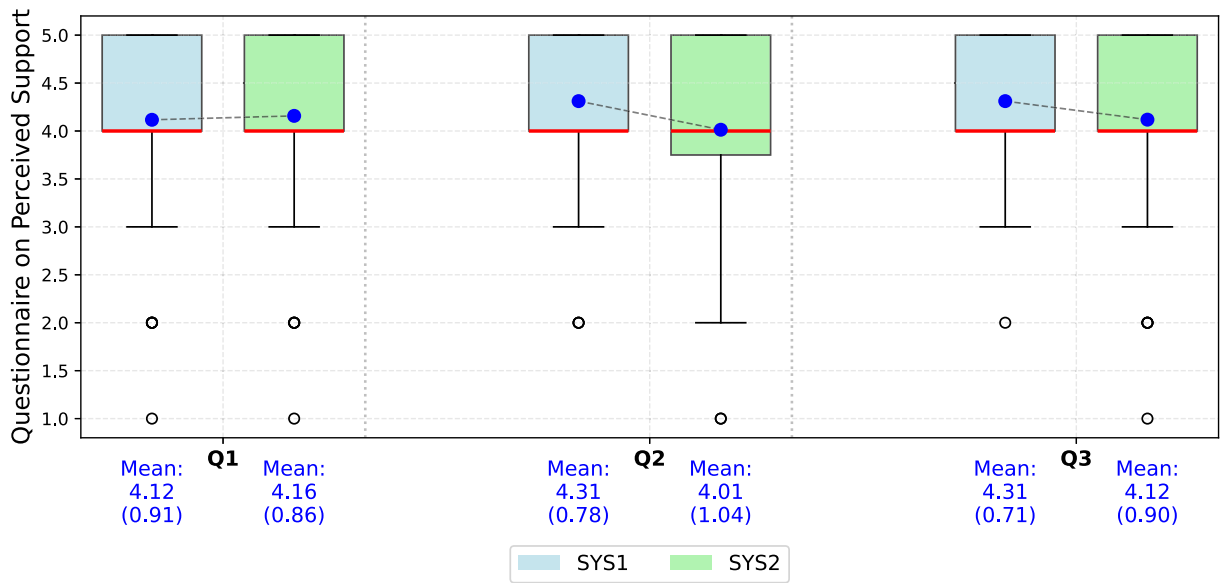


Fig. B.2. Perceived Support results for SYS1 and SYS2 across the statements of the post-task questionnaire (statements Q1, Q2, and Q3 in Table 9). We use the same notation as in Fig. B.1.

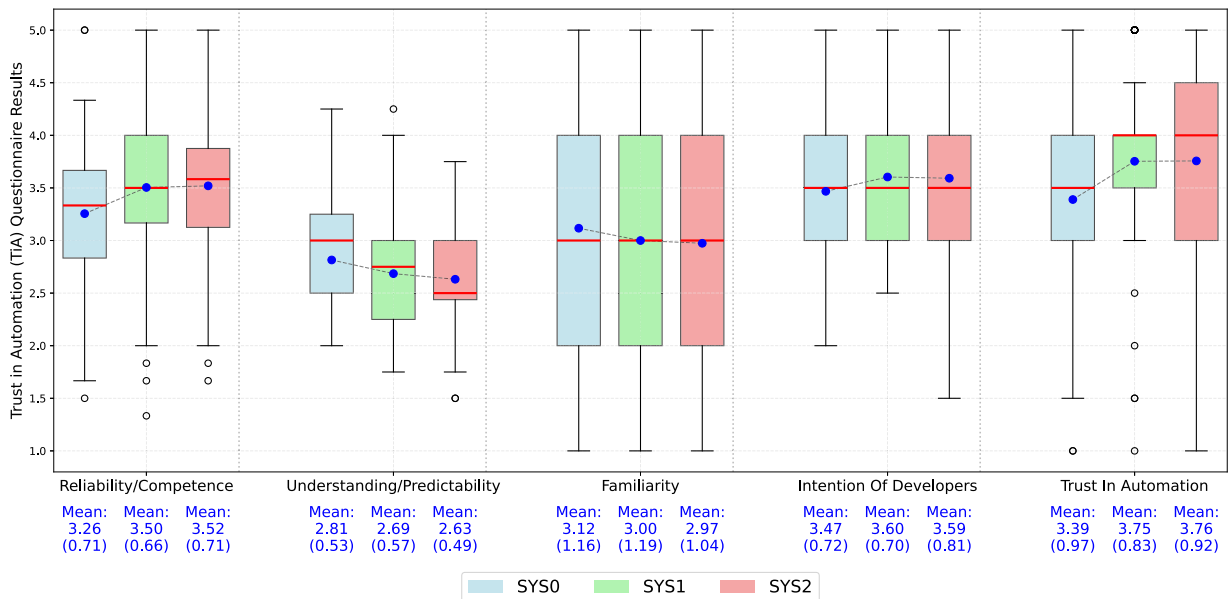


Fig. B.3. Trust in Automation (TiA) assessment results for SYS0, SYS1, and SYS2 across the five trust dimensions of Table 10 (Reliability/Competence, etc.). We use the same notation as in Fig. B.1.

Appendix C. Pairwise comparisons

Table C.1

Precision and recall pairwise comparisons: effect sizes and statistical power.

Metric	SYS0–SYS1		SYS0–SYS2		SYS1–SYS2	
	p-value	d (Power)	p-value	d (Power)	p-value	d (Power)
Precision	0.124279160	−0.636 (0.963)	1.000000000	−0.590 (0.934)	0.142022457	0.133 (0.122)
Recall	0.000000000 (***)	−1.746 (0.000)	0.000000000 (***)	−2.935 (1.000)	0.035446152 (*)	−0.519 (0.861)

Table C.2

NASA-TLX pairwise comparisons: effect sizes and statistical power.

Dimension	SYS0–SYS1		SYS0–SYS2		SYS1–SYS2	
	p-value	d (Power)	p-value	d (Power)	p-value	d (Power)
N1	0.001411475 (**)	0.677 (0.979)	0.000001105 (***)	0.981 (1.000)	0.433957152	0.262 (0.336)
N2	0.285888339	0.226 (0.265)	0.004228721 (**)	0.506 (0.843)	0.325875707	0.251 (0.314)
N3	0.004544822 (**)	0.565 (0.914)	0.000001985 (***)	0.950 (1.000)	0.144665594	0.352 (0.543)
N4	0.000031171 (***)	0.803 (0.997)	0.000000005 (***)	1.193 (1.000)	0.403822457	0.290 (0.399)
N5	0.070009712	0.479 (0.805)	0.000000464 (***)	0.991 (1.000)	0.296157806	0.380 (0.606)
N6	0.000737834 (***)	0.684 (0.981)	0.002838459 (**)	0.623 (0.955)	0.498215731	−0.071 (0.070)

Table C.3

Trust in automation (TiA) pairwise comparisons: effect sizes and statistical power.

Dimension	SYS0–SYS1		SYS0–SYS2		SYS1–SYS2	
	p-value	d (Power)	p-value	d (Power)	p-value	d (Power)
Reliability/Competence	0.246632570	−0.359 (0.562)	0.292646257	−0.368 (0.580)	1.000000000	−0.022 (0.052)
Understanding/Predictability	1.000000000	0.236 (0.285)	0.602149744	0.359 (0.557)	1.000000000	0.100 (0.091)
Familiarity	1.000000000	0.099 (0.090)	1.000000000	0.129 (0.118)	1.000000000	0.023 (0.052)
Intention of Developers	1.000000000	−0.190 (0.201)	1.000000000	−0.162 (0.158)	1.000000000	0.015 (0.051)
Trust in Automation	0.260928853	−0.402 (0.657)	0.151672924	−0.386 (0.620)	1.000000000	−0.004 (0.050)

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